Note: Content prior to 2026-03-12 was lost when a subagent overwrote this file. The literature review is preserved separately at `/tmp/literature_review.html` *. Content below was reconstructed from the 2026-03-12/13 session.*

> *Deterministic epidemic trajectory for a Gamma-distributed generation interval*

Renewal equation for arbitrary generation intervals

With a general generation interval distribution $g(\tau)$, the deterministic epidemic is governed by the **renewal equation**, not an ODE:

$$j(t) = S(t) \cdot \int_0^t R_0\, g(\tau)\, j(t-\tau)\, d\tau$$

where $j(t) = dC/dt$ is the incidence rate, $S(t) = 1 - C(t)/N$ is the susceptible fraction, and $g(\tau)$ is the generation interval density — here $\mathrm{Gamma}(\alpha_{\text{pop}},\, r)$ with $r = \alpha_{\text{pop}}/T$.

This is a **Volterra integral equation of the second kind**. There are two natural approaches for Gamma generation intervals:

Option 1: Discretised renewal equation

Discretise time with step Δt. Then at each time step i:

$$j_i = S_i \cdot R_0 \sum_{k=1}^{i} g(k\Delta t)\, j_{i-k}\, \Delta t, \qquad C_i = C_{i-1} + j_i\, \Delta t, \qquad S_i = 1 - C_i/N$$

where $g(k\Delta t) = f_{\text{Gamma}}(k\Delta t;\, \alpha_{\text{pop}},\, r)$. Seed with $j_0 = 1/N$. This is:

- **General**: works for any α_{pop} (integer or not), and indeed any g
- **Simple**: a single `for` loop in R, ~10 lines
- **Accurate**: with $\Delta t = 0.01$ it's essentially exact
- Requires $O(n^2)$ work (convolution at each step), but for $N = 1000$ and $T_{\max} \sim 100$ this is instant

Option 2: Linear chain trick (ODE, integer α_{pop} only)

When $\alpha_{\text{pop}} = n$ is a positive integer, the Gamma distribution is an Erlang, and we can convert the integral equation to an ODE system. The key observation is that for n sequential stages each with rate r:

$$P(\text{in stage } n \text{ at time } \tau) = \frac{e^{-r\tau}(r\tau)^{n-1}}{(n-1)!}$$

and the Gamma density satisfies $g(\tau) = r \cdot P(\text{in stage } n \text{ at } \tau)$. So if an individual transmits **only while in the last stage**, at rate $\beta = R_0 \cdot r$, the generation interval is exactly $\text{Gamma}(n, r)$.

The ODE system has n sequential stages $I_1, \ldots, I_n$, with transmission from I_n only:

$$\frac{dS}{dt} = -\beta\, I_n\, S$$

$$\frac{dI_1}{dt} = \beta\, I_n\, S - r\, I_1$$

$$\frac{dI_k}{dt} = r\, I_{k-1} - r\, I_k, \qquad k = 2, \ldots, n$$

$$\frac{dR}{dt} = r\, I_n$$

with $\beta = R_0 \cdot r$, initial conditions $I_1(0) = 1/N$, all others zero. This is equivalent to: stages 1 through $n-1$ are "latent" (transitioning at rate r each), and stage n is "infectious" (transmitting at rate β and recovering at rate r). The generation interval is $\text{Gamma}(n-1, r) + \text{Exp}(r) = \text{Gamma}(n, r)$. ✓

For $\alpha_{\text{pop}} = 10$, this gives a 10-compartment ODE — perfectly tractable with `odin`.

Recommendation

The **discretised renewal equation** is the better choice here because:

1. It works for any α_{pop} (you may want non-integer values later)
2. It's trivial to implement — no ODE solver or `odin` compilation needed
3. It's transparent: the generation interval $g(\tau)$ appears explicitly
4. It generalises immediately if you change the generation interval distribution

A minimal R implementation:

```
renewal_epidemic <- function(R0, popshape, T_gen, N, dt = 0.01, tmax = 100) {
  r <- popshape / T_gen
  times <- seq(0, tmax, by = dt)
  nt <- length(times)

  j <- numeric(nt)       # incidence rate
  C <- numeric(nt)       # cumulative infections
  j[1] <- 1 / N          # seed
  C[1] <- 1 / N

  # Pre-compute generation interval weights
  g <- dgamma(times, shape = popshape, rate = r) * dt

  for (i in 2:nt) {
    S <- 1 - C[i-1]
    # Convolution: sum over past incidence weighted by g
    j[i] <- S * R0 * sum(g[2:i] * j[(i-1):1])
    C[i] <- C[i-1] + j[i] * dt
  }

  tibble(t = times, j = j, cuminf = C, S = 1 - C)
}
```

> *Initial conditions for the renewal equation: matching "one person infected at t=0"*

The delta-function seed

The renewal equation for the incidence rate $j(t)$ (as a fraction of the population) seeded by a single index case at $t = 0$ is:

$$j(t) = S(t) \cdot R_0 \left[\frac{g(t)}{N} + \int_0^t g(\tau)\, j(t-\tau)\, d\tau \right]$$

The two terms have different origins:

- **Index case**: $g(t)/N$ — one person (fraction $1/N$) infected at $t = 0$ produces offspring at rate $R_0 g(\tau)$ at time τ. No $d\tau$ here because it's a single discrete person, not a continuous flow.
- **All subsequent cases**: $\int g(\tau)\, j(t-\tau)\, d\tau$ — the $j(t-\tau)\, d\tau$ fraction infected during $[t-\tau, t-\tau+d\tau]$ produces offspring via the same kernel.

What went wrong

The current code sets `j[1] = 1/N` and lets the convolution handle everything:

$$j_i = S_i \cdot R_0 \sum_k \underbrace{g(k\Delta t) \cdot \Delta t}_{=\texttt{g[k]}} \cdot\ j_{i-k}$$

When the index case enters this sum at lag $k = i$, its contribution is:

$$g(i\Delta t) \cdot \Delta t \cdot \frac{1}{N} = \frac{g(i\Delta t)}{N} \cdot \Delta t$$

But the correct contribution is $g(i\Delta t)/N$ — **without** the extra Δt. The index case is a delta function $\delta(t)/N$ in incidence, not a continuous rate $1/N$. By treating it as a rate, the seed is effectively $\Delta t = 0.01$ times too weak. Each subsequent generation inherits this factor, so the

entire trajectory is delayed by roughly $\log(1/\Delta t)/\alpha \approx 4.6$ days.

The fix

Separate the index case from the convolution explicitly:

```
renewal_epidemic <- function(R0, popshape, T_gen, N, dt = 0.01, tmax = 100) {
  r <- popshape / T_gen
  times <- seq(0, tmax, by = dt)
  nt <- length(times)

  j <- numeric(nt)
  C <- numeric(nt)
  C[1] <- 1 / N

  # Generation interval weights (with dt baked in for the convolution)
  g <- dgamma(times, shape = popshape, rate = r) * dt
  # Index case kernel (no dt — it's a single person, not a rate)
  g_index <- dgamma(times, shape = popshape, rate = r) / N

  for (i in 2:nt) {
    S <- 1 - C[i-1]
    # Index case + renewal convolution
    j[i] <- S * R0 * (g_index[i] + sum(g[2:i] * j[(i-1):1]))
    C[i] <- C[i-1] + j[i] * dt
  }

  tibble(t = times, j = j, cuminf = C, S = 1 - C)
}
```

This is equivalent to setting `j[1] = 1/(N*dt)` (the discretised delta function), but makes the reasoning explicit.

> *Why does κ=0.25 appear to be the slowest to reach 100 cases?*

The survival curves cross, but κ=0 is still slowest overall

For the Gamma model ($\alpha_{\text{pop}} = 10,\ T = 5,\ R_0 = 12$), the W distribution at each κ is approximately Gamma with shape $k = 1/\text{Var}(W)$, rate k, mean 1:

κ	Var(W)	Gamma shape	Median W	Median T_{100}	Mean T_{100}
0	0.90	1.11	0.72	8.74	9.07
0.25	0.68	1.47	0.78	8.59	8.83
1	0.16	6.31	0.95	8.26	8.31

The ordering is monotonic: $\kappa = 0$ (spike) has the latest median and mean milestone time, as expected. **But the difference between $\kappa = 0$ and $\kappa = 0.25$ is tiny — only ~0.15 days in the median.**

Why it looks reversed with 100 simulations

The milestone time has substantial stochastic spread. For a Gamma(k, k) distribution of W, the variance of $\log W$ is $\psi_1(k)$ (the trigamma function), so:

$$\text{SD}(T_{100}) \ \approx\ \frac{\sqrt{\psi_1(k)}}{\alpha}$$

This gives SD ≈ 2.2 days for $\kappa = 0$ and ≈ 1.8 days for $\kappa = 0.25$. With $n_{\text{sim}} = 100$, the standard error of the median is:

$$\text{SE}_{\text{median}} \approx \frac{\sqrt{\pi/2}}{\sqrt{n}} \cdot \text{SD} \ \approx\ 0.25 \text{ days}$$

So the SE of the *difference* in medians is ≈ 0.35 days. The true difference (0.15 days) is only **0.4 standard errors** — well within noise. The apparent reversal between $\kappa = 0$ and $\kappa = 0.25$ is a simulation artefact that will disappear with more runs.

The crossing mechanism is real, though

Your intuition about the variance effect is correct — the $\kappa = 0$ and $\kappa = 0.25$ survival curves genuinely **cross** (at $t \approx 3.5$ and $t \approx 7.8$). In the window between these crossings, $\kappa = 0$ has *lower* survival (appears faster) because its heavy right tail of W produces a handful of very fast epidemics. But this crossing isn't strong enough to make $\kappa = 0.25$ the slowest overall — it just means the two curves interleave.

The underlying reason the differences are so small: $\alpha_{\text{pop}} = 10$ makes the generation interval quite tightly concentrated (CV = 0.32), which limits how much the punctuation parameter κ can spread out W. With a less concentrated generation interval (smaller α_{pop}), the separation between κ values would be much more dramatic.

> *Gamma approximation to the SEIR generation interval (e_dur=2, i_dur=3)*

Moment-matching the SEIR generation interval

In the SEIR ODE, a person infected at time 0 spends $\text{Exp}(\nu)$ time latent, then transmits at constant rate β for an $\text{Exp}(\gamma)$ infectious period. The population-level infectiousness profile is:

$$A(\tau) = \beta \cdot P(\text{in } I \text{ at time } \tau) = \beta \cdot \frac{\nu}{\nu - \gamma}\left(e^{-\gamma\tau} - e^{-\nu\tau}\right)$$

with $\nu = 1/2,\ \gamma = 1/3$. Since $\int_0^\infty A(\tau)\,d\tau = R_0$, the generation interval density is:

$$g(\tau) = \frac{A(\tau)}{R_0} = e^{-\tau/3} - e^{-\tau/2}$$

This is a **hypoexponential** density — the distribution of $T_E + T_I$ where $T_E \sim \text{Exp}(1/2)$ and $T_I \sim \text{Exp}(1/3)$ are independent. Its moments:

$$E[\tau] = 2 + 3 = 5, \qquad \text{Var}(\tau) = 4 + 9 = 13, \qquad \text{CV} = \frac{\sqrt{13}}{5} \approx 0.72$$

Gamma moment-match

Matching $\text{Gamma}(\alpha_{\text{pop}},\ r)$ to these two moments: mean $= \alpha_{\text{pop}}/r = 5$, variance $= \alpha_{\text{pop}}/r^2 = 13$. Solving:

$$\alpha_{\text{pop}} = \frac{(\text{mean})^2}{\text{var}} = \frac{25}{13} \approx 1.92, \qquad r = \frac{\text{mean}}{\text{var}} = \frac{5}{13} \approx 0.385$$

Comparison

Model	α_{pop}	r	Mean	Variance	CV
SEIR (hypoexp)	—	—	5	13	0.72
Gamma match	1.92	0.385	5	13	0.72
Current model	10	2	5	2.5	0.32

The SEIR generation interval is **much wider** than Gamma(10, 2) — over 5× the variance. The moment-matched Gamma has shape $\alpha_{\text{pop}} \approx 1.92$, which is nowhere near 10. The SEIR model's exponentially-distributed E and I periods create a very dispersed generation interval.

This explains why punctuation effects (κ) were so dramatic in the original `1_1_basicmetrics.R` simulations (spike vs smooth vs stepwise) but so muted with $\alpha_{\text{pop}} = 10$: a wider generation interval gives κ more room to redistribute variance between the shared and individual components of the offspring timing.

> *Why does κ=0.25 look so similar to the spike (κ=0)? [Measles parameters]*

Var(W) is highly convex in κ

Your thinking is right that $\kappa = 0.25$ makes the individual infectiousness profile's SD equal to half

the population GI's SD ($\sqrt{\kappa} = 0.5$). But the **effect on Var(W) is highly nonlinear** — the extra variance beyond the smooth baseline drops like $R_0^{-2\kappa}$, which is convex and front-loaded.

The key formula

The extra variance beyond the smooth limit decomposes as:

$$\Delta(\kappa) \;=\; \mathrm{Var}(W)\big|_{\kappa} - \mathrm{Var}(W)\big|_{\kappa=1} \;=\; \frac{R_0^2}{1-m_2}\,\underbrace{\rho^{2\kappa\alpha_{\text{pop}}}}_{\text{prefactor}} \cdot \underbrace{\left(\rho_2^{(1-\kappa)\alpha_{\text{pop}}} - \rho^{2(1-\kappa)\alpha_{\text{pop}}}\right)}_{\text{bracket}}$$

Since $\rho^{\alpha_{\text{pop}}} = 1/R_0$, the prefactor simplifies to:

$$\rho^{2\kappa\alpha_{\text{pop}}} = R_0^{-2\kappa}$$

This factor is **exponential in** κ and drops very steeply near $\kappa = 0$:

κ	$R_0^{-2\kappa}$	Var(W)	% of spike extra var
0	1.000	0.798	**100%**
0.10	0.608	0.720	88%
0.25	0.289	0.608	**71%**
0.50	0.083	0.438	44%
0.75	0.024	0.286	21%
1	0.007	0.150	**0%**

At $\kappa = 0.25$, the individual profile has half the SD of the population GI — but **71% of the spike's extra variance in W is retained**. You'd need roughly $\kappa \approx 0.5$ for a halfway effect.

Intuition

The $R_0^{-2\kappa}$ prefactor comes from the Laplace transform of the jitter component evaluated at 2α. The jitter's shape is $\kappa\alpha_{\text{pop}}$, and even a small shape parameter produces substantial "diversification" — reducing variance by spreading offspring times apart. But the marginal diversification benefit

diminishes rapidly: the first 25% of jitter variance eliminates only 29% of the extra W-variance, while the last 25% eliminates 21%.

The effect is stronger at higher R_0 because $R_0^{-2\kappa}$ drops faster. For $R_0 = 2$ (influenza), $R_0^{-0.5} = 0.71$ — a modest drop. For $R_0 = 12$ (measles), $R_0^{-0.5} = 0.29$ — a steep drop but still leaving most of the extra variance. The net effect is that $\kappa = 0.25$ stays close to the spike across all realistic R_0.

> *Derivation of Var(W) = ($c^\alpha + \sigma^{(1-\psi)\alpha} - 1$) / ($1 - c^\alpha$)*

Deriving the variance of the CMJ limit W

Setup

Each infector produces $N \sim \text{Poisson}(R_0)$ offspring. Offspring j is infected at time $\tau_j = s + \varepsilon_j$, where:

- $s \sim \text{Gamma}((1-\psi)\alpha,\ r)$ — **shared shift**, common to all offspring of this infector
- $\varepsilon_j \sim \text{Gamma}(\psi\alpha,\ r)$ — **independent jitter**, iid across offspring

Marginally, $\tau_j \sim \text{Gamma}(\alpha, r)$ (by the additivity of Gamma distributions with the same rate).

The Malthusian parameter α_M satisfies the **Euler–Lotka equation**:

$$R_0 \cdot E[e^{-\alpha_M \tau}] = 1$$

Define $\rho = \frac{r}{r+\alpha_M}$, so $E[e^{-\alpha_M \tau}] = \rho^\alpha$ and the Euler–Lotka equation becomes $R_0 \rho^\alpha = 1$, i.e. $\rho^\alpha = 1/R_0$.

Step 1: The branching recursion for W

The CMJ limit $W = \lim_{t\to\infty} e^{-\alpha_M t} I_b(t)$ satisfies the distributional fixed-point equation:

$$W \stackrel{d}{=} \sum_{j=1}^{N} e^{-\alpha_M \tau_j} W_j$$

where the W_j are iid copies of W, independent of $(N, s, \varepsilon_1, \ldots)$. Define the **discounted offspring sum**:

$$S = \sum_{j=1}^{N} e^{-\alpha_M \tau_j}$$

By the Euler–Lotka equation, $E[S] = R_0 \cdot E[e^{-\alpha_M \tau}] = 1$, and from the fixed point, $E[W] = E[S] \cdot E[W] = 1$ (starting from $I_0 = 1$).

Step 2: Var(W) in terms of Var(S)

From the fixed-point equation, compute $E[W^2]$ by conditioning on $(N, s, \varepsilon_1, \ldots, \varepsilon_N)$:

$$E[W^2 \mid N, s, \boldsymbol{\varepsilon}] = \sum_j e^{-2\alpha_M \tau_j} E[W^2] + \sum_{j\neq k} e^{-\alpha_M(\tau_j+\tau_k)}$$

Recognising that $S^2 = \sum_j e^{-2\alpha_M \tau_j} + \sum_{j\neq k} e^{-\alpha_M(\tau_j+\tau_k)}$, this becomes:

$$E[W^2 \mid N, s, \boldsymbol{\varepsilon}] = (E[W^2] - 1) \sum_j e^{-2\alpha_M \tau_j} + S^2$$

Taking expectations and defining $m_2 = E\left[\sum_j e^{-2\alpha_M \tau_j}\right] = R_0 \rho_2^{\alpha}$ where $\rho_2 = \frac{r}{r+2\alpha_M}$:

$$E[W^2] = (E[W^2] - 1)\, m_2 + E[S^2]$$

Rearranging, and using $E[W] = E[S] = 1$:

> $$\boxed{\mathrm{Var}(W) = \frac{\mathrm{Var}(S)}{1 - m_2}}$$

This is valid whenever $m_2 < 1$, which holds in the supercritical regime.

Step 3: Computing E[S²] by conditioning on (N, s)

Factor the offspring times as $\tau_j = s + \varepsilon_j$:

$$S = e^{-\alpha_M s} \sum_{j=1}^{N} e^{-\alpha_M \varepsilon_j}$$

$$S^2 = e^{-2\alpha_M s} \left(\sum_{j=1}^{N} e^{-\alpha_M \varepsilon_j} \right)^2$$

Condition on N and s. The ε_j are iid $\mathrm{Gamma}(\psi\alpha, r)$, so:

$$E[e^{-\alpha_M \varepsilon}] = \rho^{\psi\alpha} \tag{1}$$

$$E[e^{-2\alpha_M \varepsilon}] = \rho_2^{\psi\alpha} \tag{2}$$

Expanding the square:

$$E\left[\left(\sum_j e^{-\alpha_M \varepsilon_j} \right)^2 \middle| N \right] = N\, \rho_2^{\psi\alpha} + N(N-1)\, \rho^{2\psi\alpha}$$

Taking expectation over $N \sim \mathrm{Poisson}(R_0)$ (using $E[N] = R_0$, $E[N(N-1)] = R_0^2$):

$$E\left[\left(\sum_j e^{-\alpha_M \varepsilon_j} \right)^2 \right] = R_0\, \rho_2^{\psi\alpha} + R_0^2\, \rho^{2\psi\alpha}$$

Now take expectation over $s \sim \mathrm{Gamma}((1-\psi)\alpha, r)$, using $E[e^{-2\alpha_M s}] = \rho_2^{(1-\psi)\alpha}$:

$$E[S^2] = \rho_2^{(1-\psi)\alpha} \left[R_0\, \rho_2^{\psi\alpha} + R_0^2\, \rho^{2\psi\alpha} \right] = R_0\, \rho_2^{\alpha} + R_0^2\, \rho^{2\psi\alpha}\, \rho_2^{(1-\psi)\alpha}$$

Since $E[S] = 1$:

$$\mathrm{Var}(S) = R_0\, \rho_2^{\alpha} + R_0^2\, \rho^{2\psi\alpha}\, \rho_2^{(1-\psi)\alpha} - 1$$

Step 4: Introducing c and σ

Define $\mu = R_0^{1/\alpha}$, the **per-stage growth factor**. From $\rho^\alpha = 1/R_0$, we get $\rho = 1/\mu$, so $\alpha_M = (\mu - 1)r$. Then:

$$\rho_2 = \frac{r}{r + 2\alpha_M} = \frac{r}{(2\mu - 1)r} = \frac{1}{2\mu - 1}$$

Define two key ratios:

> $$c = \frac{\rho_2}{\rho} = \frac{\mu}{2\mu - 1}, \qquad \sigma = \frac{\rho_2}{\rho^2} = \frac{\mu^2}{2\mu - 1} = \mu \cdot c$$

Both depend only on $\mu = R_0^{1/\alpha}$, not on ψ.

Express the terms:

First term: $m_2 = R_0\, \rho_2^\alpha = \rho_2^\alpha / \rho^\alpha = (\rho_2/\rho)^\alpha = c^\alpha$.

Second term: Using $\rho^{2\psi\alpha} = (\rho^\alpha)^{2\psi} = R_0^{-2\psi}$:

$$R_0^2\, \rho^{2\psi\alpha}\, \rho_2^{(1-\psi)\alpha} = R_0^{2(1-\psi)}\, \rho_2^{(1-\psi)\alpha} = \left(\frac{\rho_2}{\rho^2}\right)^{(1-\psi)\alpha} = \sigma^{(1-\psi)\alpha}$$

where we used $\rho^{2(1-\psi)\alpha} = R_0^{-2(1-\psi)}$ to write $R_0^{2(1-\psi)}\, \rho_2^{(1-\psi)\alpha} = (\rho_2/\rho^2)^{(1-\psi)\alpha}$.

Step 5: Final result

Substituting into $\mathrm{Var}(W) = \mathrm{Var}(S)/(1 - m_2)$:

> $$\boxed{\mathrm{Var}(W) = \frac{c^\alpha + \sigma^{(1-\psi)\alpha} - 1}{1 - c^\alpha}}$$

where $c = \frac{\mu}{2\mu-1}$, $\sigma = \frac{\mu^2}{2\mu-1}$, $\mu = R_0^{1/\alpha}$.

Key observations

- **Smooth limit** ($\psi = 1$): $\sigma^0 = 1$, so $\mathrm{Var}(W) = c^\alpha/(1-c^\alpha) = m_2/(1-m_2)$. This is the **minimum** variance — each offspring time is fully independent.
- **Spike limit** ($\psi = 0$): $\sigma^\alpha = R_0^2\rho_2^\alpha$, giving $\mathrm{Var}(W) = (c^\alpha + \sigma^\alpha - 1)/(1-c^\alpha)$. This is the **maximum** — all offspring share the same timing.
- ψ only enters through $\sigma^{(1-\psi)\alpha}$, confirming that narrower profiles (smaller ψ) always increase $\mathrm{Var}(W)$.
- Since $E[W] = 1$, we have $\mathrm{CV}^2(W) = \mathrm{Var}(W)$.

> *Conditional moments, Gamma approximation, and time-shift mode*

From Var(W) to the mode of the time shift

Conditional moments of W

The random variable W has a point mass at 0 (extinction, probability q) and a continuous part on $(0, \infty)$ (survival, probability $1-q$). Since all unconditional moments of W come entirely from the continuous part:

$$E[W^n] = (1-q)\, E[W^n \mid W > 0]$$

so the conditional (on survival) raw moments are simply:

$$E[W^n \mid \mathrm{surv}] = \frac{E[W^n]}{1-q}$$

For the extinction probability, q is the smallest root of $q = e^{R_0(q-1)}$ (Poisson offspring). For

$R_0 \gg 1, q \approx 0.$

We already derived $E[W] = 1$ and:

$$E[W^2] = 1 + \mathrm{Var}(W) = 1 + \frac{c^\alpha + \sigma^{(1-\psi)\alpha} - 1}{1 - c^\alpha} = \frac{\sigma^{(1-\psi)\alpha}}{1 - c^\alpha}$$

where we combined terms: $1 + (c^\alpha + \sigma^{(1-\psi)\alpha} - 1)/(1 - c^\alpha) = (1 - c^\alpha + c^\alpha + \sigma^{(1-\psi)\alpha} - 1)/(1 - c^\alpha)$.

The conditional coefficient of variation squared is:

$$\begin{aligned} \mathrm{CV}^2_{\mathrm{cond}} &= \frac{\mathrm{Var}(W \mid \mathrm{surv})}{(E[W \mid \mathrm{surv}])^2} && (3) \\ &= \frac{E[W^2 \mid \mathrm{surv}]}{(E[W \mid \mathrm{surv}])^2} - 1 && (4) \\ &= \frac{E[W^2]/(1-q)}{1/(1-q)^2} - 1 && (5) \\ &= (1-q)\, E[W^2] - 1 && (6) \end{aligned}$$

Substituting $E[W^2] = \sigma^{(1-\psi)\alpha}/(1 - c^\alpha)$:

> $$\mathrm{CV}^2_{\mathrm{cond}} = \frac{(1-q)\,\sigma^{(1-\psi)\alpha}}{1 - c^\alpha} - 1$$

When $q \approx 0$: $\mathrm{CV}^2_{\mathrm{cond}} \approx \mathrm{Var}(W)$, since $E[W] = 1$ makes $\mathrm{CV}^2 = \mathrm{Var}$ unconditionally.

Two-moment Gamma approximation

Approximate $W \mid \mathrm{surv}$ by $\mathrm{Gamma}(k, \lambda)$, matching the first two moments. Write $V = \mathrm{CV}^2_{\mathrm{cond}}$

and $p = 1 - q$ for brevity.

Quantity	In terms of k, λ	Solved
Mean	$k/\lambda = 1/p$	$k = 1/V$
CV²	$1/k = V$	$\lambda = p/V$

For $\mathrm{Gamma}(k, \lambda)$, the n-th raw moment is $\prod_{j=0}^{n-1}(k+j)/\lambda^n$. Using $k = 1/V$ and $\lambda = p/V$:

$$E[W \mid \mathrm{surv}] = \frac{1}{p} \tag{7}$$

$$E[W^2 \mid \mathrm{surv}] = \frac{1+V}{p^2} \tag{8}$$

$$E[W^3 \mid \mathrm{surv}] = \frac{(1+V)(1+2V)}{p^3} \tag{9}$$

The first two are exact (matched by construction). The third is a **prediction** of the Gamma approximation — it will differ from the true $E[W^3 \mid \mathrm{surv}]$ unless $W \mid \mathrm{surv}$ happens to be exactly Gamma.

Expanding V gives the fully explicit forms. For instance:

$$E[W^2 \mid \mathrm{surv}] = \frac{1}{p^2}\left(1 + \frac{p\,\sigma^{(1-\psi)\alpha}}{1-c^\alpha} - 1\right) + \frac{1}{p^2} = \frac{\sigma^{(1-\psi)\alpha}}{p(1-c^\alpha)}$$

which matches the exact value $E[W^2]/(1-q) = \sigma^{(1-\psi)\alpha}/[p(1-c^\alpha)]$. ✓

Mode of the time shift

The time shift is $\varsigma = \log(W)/\alpha_M$ (written as $\log(W)/r$ in the writeup, where r denotes the growth rate α_M). Its mode is determined by the mode of $W \mid \mathrm{surv}$.

The mode of a $\mathrm{Gamma}(k, \lambda)$ is $(k-1)/\lambda$ (for $k \geq 1$). Substituting:

$$\mathrm{mode}(W \mid \mathrm{surv}) = \frac{k-1}{\lambda} = \frac{1/V - 1}{p/V} = \frac{1-V}{p}$$

Taking the log and dividing by α_M:

$$\mathrm{mode}_\varsigma = \frac{1}{\alpha_M}\left[\log(1-V) - \log p\right] = \frac{1}{\alpha_M}\left[\log(1-\mathrm{CV}^2_{\mathrm{cond}}) - \log(1-q)\right]$$

When extinction is negligible ($q \approx 0$, so $p \approx 1$):

> $$\boxed{\mathrm{mode}_\varsigma \approx \frac{1}{\alpha_M}\log\left(1-\mathrm{CV}^2_{\mathrm{cond}}\right)}$$

which is exactly the expression in the writeup (with the identification $r \equiv \alpha_M$).

Interpretation

- **Sign:** Since $V = \mathrm{CV}^2_{\mathrm{cond}} > 0$, the mode is $\log(1-V) < 0$, hence $\mathrm{mode}_\varsigma < 0$. The most likely epidemic is **delayed** relative to the deterministic trajectory — the mode of W is less than its mean.
- **Magnitude:** Larger V (more punctuated profile, smaller ψ) pushes the mode further negative — the typical epidemic starts later. In the extreme, $V \to 1$ makes $\mathrm{mode}_\varsigma \to -\infty$.
- **ψ-dependence:** Substituting $V \approx \mathrm{Var}(W) = (c^\alpha + \sigma^{(1-\psi)\alpha} - 1)/(1-c^\alpha)$, the mode depends on ψ solely through $\sigma^{(1-\psi)\alpha}$. The $\psi = 1$ (smooth) limit gives the least negative mode; $\psi = 0$ (spike) gives the most negative.
- **Skewness:** The Gamma approximation has skewness $2/\sqrt{k} = 2\sqrt{V}$. More punctuated profiles increase V, making W more right-skewed: a few epidemics start very early, but the bulk are delayed.

Validity constraint

The mode formula requires $k \geq 1$, i.e. $V \leq 1$, i.e. $\mathrm{CV}^2_{\mathrm{cond}} \leq 1$. When $V > 1$, the Gamma density is monotonically decreasing (mode at 0), and the mode of the time shift is $-\infty$ — the most likely outcome becomes extinction or near-extinction. This occurs for very punctuated profiles with moderate R_0. For the spike limit ($\psi = 0$) with $q \approx 0$:

$$V_{\mathrm{spike}} = \frac{c^\alpha + \sigma^\alpha - 1}{1 - c^\alpha} = \frac{c^\alpha + R_0^2 \rho_2^\alpha - 1}{1 - c^\alpha}$$

This exceeds 1 when $\sigma^\alpha > 2(1 - c^\alpha)$, which can happen for small R_0 or small α.

> *Overdispersion from periodic contacts × punctuated infectiousness*

Deriving the dispersion parameter k

Setup

Consider a population with **periodic contact rate**:

$$c(t) = \bar{c} + A \sin(\omega t)$$

where $\bar{c}$ is the mean contact rate, $A < \bar{c}$ is the amplitude, and $\omega = 2\pi/P$ is the angular frequency (period P). An individual infected at calendar time t_0 has a biological infectiousness profile $a_i(\tau)$ — a density over time-since-infection τ. Their individual reproduction number is:

$$\mathcal{R}_i = R_0 \int_0^\infty a_i(\tau) \, \frac{c(t_0 + \tau)}{\bar{c}} \, d\tau$$

In the punctuated model, $a_i(\tau)$ is the density of $s_i + \varepsilon$ where $\varepsilon \sim \mathrm{Gamma}(\psi\alpha, r)$, centred on the individual's shared shift s_i. The jitter ε spreads the profile, while s_i locates it in time.

Step 1: Characteristic function argument

Substitute $c(t_0+\tau)/\bar{c} = 1 + (A/\bar{c})\sin(\omega(t_0+\tau))$:

$$\mathcal{R}_i = R_0\left[1 + \frac{A}{\bar{c}}\int_0^\infty a_i(\tau)\sin(\omega(t_0+\tau))\,d\tau\right]$$

The integral is the imaginary part of the characteristic function of a_i at frequency ω. Since a_i is the density of $s_i + \varepsilon$:

$$\int a_i(\tau)\,e^{i\omega\tau}\,d\tau = e^{i\omega s_i}\cdot\underbrace{\left(\frac{r}{r-i\omega}\right)^{\psi\alpha}}_{\text{char. fn. of }\varepsilon}$$

Write the Gamma characteristic function in polar form:

$$\left(\frac{r}{r-i\omega}\right)^{\psi\alpha} = D\,e^{i\varphi}$$

where the **damping factor** is:

$$D = \left|\frac{r}{r-i\omega}\right|^{\psi\alpha} = \left(1+\frac{\omega^2}{r^2}\right)^{-\psi\alpha/2}$$

and $\varphi = \psi\alpha\cdot\arctan(\omega/r)$ is a phase shift. Then:

$$\int a_i(\tau)\sin(\omega(t_0+\tau))\,d\tau = D\cdot\sin(\omega(t_0+s_i)+\varphi)$$

So each individual's reproduction number is:

$$\mathcal{R}_i = R_0\left[1 + \frac{A}{\bar{c}}\,D\,\sin\big(\omega(t_0+s_i)+\varphi\big)\right]$$

Step 2: Variance of $\mathcal{R}_i$

The phase $\omega(t_0+s_i)+\varphi$ varies across individuals (through both infection time t_0 and shared shift s_i). Averaging over the population — either because t_0 is uniformly distributed over the cycle,

or because s_i has enough spread — $\sin^2$ averages to 1/2:

$$E[\mathcal{R}_i] = R_0, \qquad \mathrm{Var}(\mathcal{R}_i) = R_0^2 \cdot \frac{A^2}{\bar{c}^2} \cdot D^2 \cdot \frac{1}{2}$$

Step 3: The dispersion parameter

Each individual produces $N_i \sim \mathrm{Poisson}(\mathcal{R}_i)$ offspring, so:

$$\mathrm{Var}(N_i) = E[\mathcal{R}_i] + \mathrm{Var}(\mathcal{R}_i) = R_0 + \frac{R_0^2 A^2 D^2}{2\bar{c}^2}$$

Matching to the negative binomial mean-variance relation $\mathrm{Var} = \mu + \mu^2/k$:

$$k = \frac{R_0^2}{\mathrm{Var}(\mathcal{R}_i)} = \frac{2\bar{c}^2}{A^2 D^2}$$

Substituting $D^2 = (1 + \omega^2/r^2)^{-\psi\alpha}$:

> $$\boxed{k = \frac{2\bar{c}^2}{A^2}\left(1 + \frac{\omega^2}{r^2}\right)^{\psi\alpha}}$$

Here $r = \alpha/T$ is the **Gamma rate parameter** of the generation interval (not the epidemic growth rate).

Note: The expression in `uncontrolled.tex` line 37 has a LaTeX typo: `\frac{2}{\bar{c}^2}{A^2}` renders as $2A^2/\bar{c}^2$, but the correct coefficient is $2\bar{c}^2/A^2$. (Sanity check: $k \to \infty$ as $A \to 0$, recovering Poisson.)

Step 4: The critical period

Define the **critical period**:

$$z^*_{\mathrm{per}} = \frac{2\pi}{r} = \frac{2\pi T}{\alpha}$$

This is the period of a contact cycle whose angular frequency matches the Gamma rate r. Since $\omega = 2\pi / z_{\text{per}}$ where $z_{\text{per}} = P$ is the contact period:

$$\frac{\omega}{r} = \frac{z^*_{\text{per}}}{z_{\text{per}}}$$

and the formula becomes:

> $$k = \frac{2\bar{c}^2}{A^2}\left(1 + \left(\frac{z^*_{\text{per}}}{z_{\text{per}}}\right)^2\right)^{\psi\alpha}$$

Three regimes

Regime	Condition	k	Interpretation
Slow cycle	$z_{\text{per}} \gg z^*_{\text{per}}$	$\approx 2\bar{c}^2/A^2$	Contact cycle so slow that even jittered offspring see a nearly constant c. Overdispersion maximal, **independent of** ψ.
Critical	$z_{\text{per}} \approx z^*_{\text{per}}$	$\approx (2\bar{c}^2/A^2) \cdot 2^{\psi\alpha}$	Maximum sensitivity to ψ: smooth profiles average out fluctuations, punctuated profiles don't.
Fast cycle	$z_{\text{per}} \ll z^*_{\text{per}}$	$\sim (z^*_{\text{per}}/z_{\text{per}})^{2\psi\alpha} \to \infty$	Offspring average over many cycles → Poisson. **Unless** $\psi = 0$ (spike), where $k = 2\bar{c}^2/A^2$ always.

Physical intuition

The key insight is that the **shared shift s does not reduce overdispersion** — it merely relocates the burst to a different phase of the contact cycle, but all offspring still see the same

contact rate. Only the **jitter** ε (controlled by ψ) averages over the contact fluctuations, because it spreads offspring across different phases.

- **Spike** ($\psi = 0$): No jitter, $D = 1$. All offspring hit the contact cycle at the same phase. $k = 2\bar{c}^2/A^2$, maximal overdispersion regardless of period.
- **Smooth** ($\psi = 1$): Full jitter with shape α. $D = (1 + \omega^2/r^2)^{-\alpha/2}$, strong damping of contact fluctuations. k grows as $(z^*_{\text{per}}/z_{\text{per}})^{2\alpha}$ for fast cycles.
- **Intermediate** ψ: Jitter shape is $\psi\alpha$, giving partial damping. The exponent $\psi\alpha$ interpolates between the two limits.

Numerical examples

For measles ($\alpha = 11.36, T = 12.2$ days): $z^*_{\text{per}} = 2\pi \cdot 12.2/11.36 \approx 6.7$ days.

For influenza ($\alpha = 2.32, T = 3.2$ days): $z^*_{\text{per}} = 2\pi \cdot 3.2/2.32 \approx 8.7$ days.

Both are near one week — so **weekly contact patterns** sit squarely in the critical regime where the punctuation parameter ψ has maximal effect on overdispersion.

> *Detailed summary of Middleton & Larremore (2024), "Modeling the transmission mitigation impact of testing for infectious diseases," Science Advances 10, eadk5108.*

Middleton & Larremore (2024): Mathematical framework

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1. Core quantity: Testing Effectiveness (TE)

The central quantity is **Testing Effectiveness (TE)**, defined in direct analogy to vaccine effectiveness (VE):

$$TE = 1 - \frac{R_{\text{testing}}}{R_{\text{no testing}}} \qquad \Longleftrightarrow \qquad R_{\text{testing}} = (1 - TE)\, R_{\text{no testing}} \tag{1}$$

TE is the proportion by which a testing program reduces population-level transmission, in expectation. It incorporates the test itself, its administration strategy, human behaviors (compliance, participation), and the interplay with viral kinetics. They note this mirrors VE: the expected reduction in post-infection transmission risk for tested vs. untested populations.

2. Infectiousness curve $\beta(t)$ and the reproduction number

Each individual has a time-varying **infectiousness curve** $\beta(t)$, where t is time since exposure. The **individual reproduction number** (in the absence of testing) is:

$$\nu_0 = \int_0^\infty \beta(t)\, dt \tag{2}$$

The population-level basic reproduction number is the mean across all individuals:

$$R_{\text{no testing}} = \mathbb{E}[\nu_0] = \int_0^\infty \mathbb{E}[\beta(t)]\, dt \tag{3}$$

The key modeling assumption is that **infectiousness is proportional to the logarithm of pathogen load above an empirically established threshold**. That is, $\beta(t) \propto \max(0,\, \log_{10} V(t) - \log_{10} V_{\text{thresh}})$, where $V(t)$ is the viral load at time t. This is consistent with observations that higher viral loads are associated with more efficient transmission for SARS, SARS-CoV-2, influenza, and RSV.

3. Viral kinetics model

Middleton & Larremore use a **piecewise linear model of log pathogen load** (a tent function on the log scale), specified by four parameters per infection trajectory:

- **Latent period**: t_0 to $t_0 + \text{LLOQ}$ -- time from exposure to when the virus first exceeds the

lower limit of quantification. During this time, $D(t) = 0$ and $\beta(t) = 0$.

- **Proliferation phase**: Exponential growth (linear increase in log VL) at some proliferation rate, from first detectability to peak VL at time t_{peak}.
- **Peak viral load**: V_{peak}, drawn from a distribution (Lognormal).
- **Clearance phase**: Exponential decline (linear decrease in log VL) from peak back down to undetectable.

All four control points (t_0, LLOQ), $(t_{\text{peak}}, V_{\text{peak}})$, and (t_f, LLOQ) are drawn from distributions (Lognormal with parameters μ, σ) estimated from the literature for each pathogen (RSV, influenza A, SARS-CoV-2 founder/naive and omicron/experienced). This yields stochastic, individual-level VL trajectories.

Symptom onset t_{Sx} is drawn relative to peak pathogen load. For example, for founder-strain SARS-CoV-2 in naive hosts, symptoms occur 0 to 3 days after peak VL; for omicron in experienced hosts, 1 to 5 days before peak VL. This reflects the typical manner of reporting in the literature. Asymptomatic infections correspond to trajectories with no symptom onset time drawn.

4. Isolation after diagnosis

If an individual receives a diagnosis at time t_{Dx}, they isolate and thereby attenuate transmission. For **perfect isolation of fixed duration** ℓ, the individual exits isolation at $t_{\text{exit}} = t_{\text{Dx}} + \ell$. The post-testing reproduction number becomes:

$$\nu_{\text{testing}}(t_{\text{Dx}}, t_{\text{exit}}) = \int_0^{t_{\text{Dx}}} \beta(t)\, dt + \int_{t_{\text{exit}}}^{\infty} \beta(t)\, dt \tag{5}$$

This has the intuitive interpretation: transmission that occurred before diagnosis plus transmission that occurs after exiting isolation. Transmission during isolation is assumed to be zero (perfect isolation). They note that **imperfect/partial isolation is developed in Supplementary Materials**, where time-varying impacts of behavioral changes (masking, increased ventilation) can

be modeled instead of binary on/off isolation.

5. Diagnosis time as a random variable

The time of diagnosis t_{Dx} depends on test availability, schedule, symptom timing, and compliance. It is modeled as a **random variable** with (improper) probability density $f(t_{\mathrm{Dx}})$. This leads to an effective infectiousness in the testing scenario, averaged over possible diagnosis times:

$$\bar{\nu}_{\text{testing}} = \int_0^\infty \nu_{\text{testing}}(t_{\mathrm{Dx}})\, f(t_{\mathrm{Dx}})\, dt_{\mathrm{Dx}} \tag{6}$$

Substituting and rearranging using the cumulative probability of diagnosis $F(t)$:

> $$\bar{\nu}_{\text{testing}} = \nu_0 - \int_0^\infty \beta(t)[F(t) - F(t-\ell)]\ dt \tag{7}$$

The term $F(t) - F(t-\ell)$ is the probability of being in isolation at time t: diagnosed before t but not yet released (released ℓ days after diagnosis). Note that $F(t)$ is an **improper CDF** -- it need not reach 1 as $t \to \infty$, because not all individuals will necessarily be diagnosed.

They observe that if $F(t)$ is reinterpreted as the CDF of symptom onset (symptoms as a perfect diagnostic test), with exhaustive isolation thereafter (large ℓ), Eq. 7 recovers the core notion of **symptom-based controllability** from Fraser et al. (2004).

6. Participation and the effective reproduction number

Let p be the proportion of the population participating in the testing program. Participants have their infectiousness reduced from ν_0 to $\bar{\nu}_{\text{testing}}$; non-participants are unaffected. The effective reproduction number is:

> $$R_{\text{testing}} = p\,\mathbb{E}[\bar{\nu}_{\text{testing}}] + (1-p)\,\mathbb{E}[\nu_0] \tag{4}$$

where the expectations are over the heterogeneous within-host kinetics across the population.

7. Calculating the diagnosis CDF $F(t)$: testing, compliance, detectability, and TAT

The CDF $F(t)$ -- the probability of having received a positive diagnosis by time t -- is built from three components:

$$A(t)$$

$A(t)$ is the prescribed rate of test administration in tests per day. For sufficiently small δ:

$$\Pr(\text{test administered between } t \text{ and } t+\delta) = \int_t^{t+\delta} A(t')\,dt' \tag{8}$$

Examples: $A(t) = 1/7$ models weekly screening; $A(t) = 2$ for $4 \leq t \leq 6$ represents twice-daily testing in a two-day window.

$$D(t)$$

$D(t)$ is a binary indicator: $D(t) = 1$ when the infection is analytically detectable at time t (pathogen load above the test's limit of detection, LOD), and $D(t) = 0$ otherwise. This defines the **detectability window**.

Combining administration with detectability, the cumulative number of scheduled tests with the potential to return a positive result by time t is:

$$m(t) = \int_0^{t-\mathrm{TAT}} A(t')\,D(t')\,dt' \tag{9}$$

The upper limit is shifted by **TAT (turnaround time)** -- the delay between a test's administration and the return of actionable results. This is a crucial parameter: even if a test is taken during the detectable window, the result is not available until TAT later.

- **Compliance** c: Independent Bernoulli probability that each prescribed test is actually taken.
- **Test failure** ϕ: Independent Bernoulli probability that a test fails for non-analytical reasons (e.g., poor biospecimen collection).

Combining these, the CDF of having received a positive test by time t is:

$$\begin{aligned} F(t) = 1 - [1 - \overline{m}(t)]\,[1 - c(1-\phi)]^{m(t)-\overline{m}(t)} \\ - \overline{m}(t)\,[1 - c(1-\phi)]^{m(t)-\overline{m}(t)+1} \end{aligned} \tag{10}$$

where $\overline{m}(t) = m(t) \bmod 1$. This function reflects that for a monotonically increasing number of possible positive tests $m(t)$, there are two possibilities: one may have taken either $m(t) - \overline{m}(t)$ tests (with probability $1 - \overline{m}(t)$) or $m(t) - \overline{m}(t) + 1$ tests (with probability $\overline{m}(t)$). Each such test may fail with probability ϕ or be missed with probability $1 - c$.

This is an **improper CDF** -- it need not reach 1, reflecting that not all individuals will be diagnosed.

When $A(t)D(t)$ is a constant $\bar{A}$ between times t_1 and t_2 (and 0 otherwise), the improper PDF simplifies to:

$$f(t) = \bar{A}c(1-\phi)\,[1 - c(1-\phi)]^{m(t)-\overline{m}(t)} \tag{11}$$

for $t_1 + \mathrm{TAT} \leq t \leq t_2 + \mathrm{TAT}$, and 0 otherwise.

For elective testing triggered by symptoms at time t_{Sx}, they model individuals as waiting x days after symptoms and then testing for y days at rate $\bar{A}$ per day, leading to a test administration function:

$$A(t, t_{\mathrm{Sx}}) = \bar{A}[u(t - t_{\mathrm{Sx}} - x) - u(t - t_{\mathrm{Sx}} - x - y)] \tag{12}$$

where $u(\cdot)$ is the unit step function. The detection CDF then depends on t_{Sx} and must be integrated over the t_{Sx} distribution (Eq. 13).

8. The TE formula

Substituting Eq. 7 ($\bar{\nu}_{\mathrm{testing}}$) into Eq. 4 ($R_{\mathrm{testing}}$) and then into Eq. 1 ($TE$):

> $$TE = p \int_0^\infty \mathbb{E}\{\beta(t)[F(t) - F(t - \ell)]\}\, dt \Big/ \int_0^\infty \mathbb{E}[\beta(t)]\, dt \tag{14}$$

The numerator is the expected infectiousness averted by isolation; the denominator is the total expected infectiousness without testing. Both integrands are **expected infectiousness trajectories** -- ensemble averages over the population's heterogeneous within-host kinetics.

When isolation is of sufficient duration to prevent all post-diagnosis transmission (large ℓ), this simplifies to:

> $$TE = p \int_0^\infty \mathbb{E}[\beta(t)\, F(t)]\, dt \Big/ \int_0^\infty \mathbb{E}[\beta(t)]\, dt \tag{15}$$

Note the subtle but important distinction: the numerator involves $\mathbb{E}[\beta(t)\, F(t)]$, not $\mathbb{E}[\beta(t)] \cdot \mathbb{E}[F(t)]$. The joint expectation matters because $\beta(t)$ and $F(t)$ are correlated -- both depend on the same within-host viral kinetics trajectory.

9. Ascertainment (protocol sensitivity)

The long-time limit of $F(t)$ gives the probability that an individual is ever diagnosed. **Ascertainment** α (also called "protocol sensitivity" in the literature) is the population-level

fraction of infections diagnosed:

$$\alpha = p \cdot \lim_{t \to \infty} \mathbb{E}[F(t)] \tag{16}$$

A key empirical finding of the paper is that **ascertainment and TE are only weakly related**. For instance, for influenza A with elective RDT testing after symptoms, TE = 38% but ascertainment = 60%; for elective RT-qPCR testing, TE dropped by 36 percentage points (to 2%) while ascertainment dropped by only 10 percentage points (to 50%). This is because TE depends on *when* diagnoses occur relative to the infectiousness curve, not just *whether* they occur.

10. Mean generation interval and selection coefficient

The mean generation interval can be computed from the centers of mass of the expected infectiousness curves. The integrands in Eq. 14 represent the expected infectiousness trajectories with (numerator) and without (denominator) testing. These estimates may be important for post hoc estimation of TE from empirical data, since methods for estimating R from case counts often rely on the mean generation interval.

If a single test can diagnose two strains of the same pathogen, testing can alter the selection coefficient. The selection coefficient shifts from $s = R_A/R_B$ to $s_{\text{testing}} = s \cdot (1 - TE_A)/(1 - TE_B)$.

11. Test consumption

They estimate two quantities for test consumption:

If no diagnosis is ever made, the expected number of tests consumed is a weighted combination of tests taken during the detectable window (each at rate $c\phi/(1 - c + c\phi)$, reflecting the probability that a planned test is consumed given absence of diagnosis) and during the non-detectable window (each at rate c):

$$\bar{q}_{\text{no-Dx}} = \frac{c\phi}{1 - c + c\phi} \int_0^\infty A(t)D(t)\,dt + c\int_0^\infty A(t)[1 - D(t)]\,dt \tag{17}$$

For a fixed time of diagnosis t_{Dx}, the expected tests consumed is 1 (the diagnosing test) plus any tests administered before it:

$$\bar{q}_{\text{Dx}}(t_{\text{Dx}}) = 1 + c\left[\left|\int_0^{t_{\text{Dx}}-\text{TAT}} A(t)\,dt\right| + \left|\int_{t_{\text{Dx}}-\text{TAT}}^{t_{\text{Dx}}} A(t)\,dt\right|\right] \tag{18}$$

The total expected test consumption per infected individual is then obtained by combining these, weighted by ascertainment. The overall population test consumption rate (including non-infected individuals) is:

$$Q = p[(\theta\, se_{\text{ref.}})\,\bar{q}_{\text{infected}} + (1-\theta)(1 - sp_{\text{ref.}})\,\bar{q}_{\text{not infected}}] \tag{22}$$

where θ is the prevalence, $se_{\text{ref.}}$ and $sp_{\text{ref.}}$ are sensitivity and specificity of the referral scheme (e.g., contact tracing, symptom presence), and $\bar{q}_{\text{not infected}} = c\int_0^\infty A(t)\,dt$.

12. Days spent in isolation

For fixed-duration perfect isolation of length ℓ, the average isolation days per diagnosed individual is simply $d = \ell \cdot \alpha$. They also model **test-to-exit (TTE)** strategies, where individuals test daily (using the same or a different test) after a minimum waiting period in isolation, and exit upon a negative test result. This is covered in the Supplementary Materials (Eq. S10).

13. Imperfect isolation and behavioral changes

While the main text develops perfect isolation (zero transmission during isolation), the Supplementary Materials provide equations for **partial and time-varying impacts of behavioral changes** such as masking or increased ventilation. The general form replaces the binary "in/out of isolation" with a time-varying infectiousness attenuation factor during the isolation period.

14. Parameterization of viral kinetics

The viral kinetics model uses **Lognormal distributions** for peak VL, proliferation duration, and clearance duration, with parameters μ and σ adapted from published means, 95% credible intervals, and sample sizes. Specifically:

- **RSV**: Challenge studies (Refs 28, 29). 2-4 day latent, 3-6 day clearance. VL between 4.5 and 8 $\log_{10}$ cp RNA/ml.
- **Influenza A**: Challenge studies (Refs 30-32). 0.5-2 day latent. Peak VL at 1-3 days after symptom onset. VL between 6 and 8.5 $\log_{10}$ cp RNA/ml.
- **SARS-CoV-2 (founder/naive)**: Prospective longitudinal studies (Refs 17, 34). Higher peaks, longer detectability windows (~6.6 days RDT detection).
- **SARS-CoV-2 (omicron/experienced)**: Lower peaks, shorter clearance, shorter detectability windows (~2.2 days RDT detection), earlier symptom onset relative to peak VL.

Because within-host kinetics vary from one infection to another, each trajectory is parameterized using **independent draws from random variables** for each control point. They use 10^5 stochastic realizations for Monte Carlo estimation.

15. Analytical sensitivity and LOD

LODs are drawn from the literature for RT-qPCR and RDT tests for each pathogen. Because of variability across assays, a **single LOD per pathogen and test type** is used. Test failure rates ϕ were available only for SARS-CoV-2 (~5%); the same rate was assumed for RSV and influenza A to model sample contamination, poor biospecimen collection, or manufacturing error.

16. Key findings and their mathematical basis

- **TE varies considerably by pathogen and strategy**: Weekly RDT screening with 50% compliance gave TE of 13% (RSV), 7% (influenza A), and 13% (SARS-CoV-2 omicron). Elective RDT at symptom onset: 21% (RSV), 38% (influenza A), 25% (SARS-CoV-2).
- **Ascertainment $\neq$ TE**: Ascertainment was only weakly correlated with TE across scenarios (11-14% ascertainment for weekly screening vs. up to 60% for elective post-symptom testing,

but TE did not track proportionally).

- **RT-qPCR vs RDT reversal**: For founder-strain/naive SARS-CoV-2, RDTs provided higher TE than RT-qPCR (replicating earlier literature). For omicron/experienced hosts, this *reversed* -- RT-qPCR gave higher TE despite its 2-day TAT, due to shortened RDT detectability windows and the larger gap between PCR and RDT detectability.
- **Presymptomatic transmission limits TE**: The existence of asymptomatic and presymptomatic transmission implies TE < 1 for symptom-driven testing, regardless of test quality.

17. Relationship to the Fraser et al. (2004) controllability framework

Eq. 7 with $F(t)$ reinterpreted as the CDF of symptom onset and $\ell \to \infty$ recovers the Fraser et al. notion of symptom-based controllability. However, Middleton & Larremore generalize this by:

1. Making t_{Dx} depend on the test's properties and within-host kinetics (not just symptom onset)
2. Including compliance, test failure, and turnaround time
3. Allowing finite isolation duration
4. Modeling heterogeneous viral kinetics across individuals

18. Computational approach

All calculations are performed either by **integration** (when $A(t)D(t)$ is a constant, leading to closed-form expressions) or by **Monte Carlo simulation** (10^5 stochastic realizations of viral load trajectories, symptom onset, test timing, and diagnosis). The ensemble mean of the infectiousness curves with and without testing (Fig. 1B) directly yields the TE estimate. They note this is "similar in spirit to the CEPAC model" for HIV, which provides stochastic individual-level realizations of infection dynamics, costs, and outcomes.

> *Formalizing how ψ enters screening effectiveness — the Gamma convolution invariance*

The Gamma convolution invariance and its breaking

The invariance theorem

Theorem. If individual i is detected at a time τ_{det} that is independent of (l_i, ε_j) — e.g., at a fixed time after infection — then with perfect immediate isolation ($\eta = 1$, $\delta_{\text{act}} = 0$):

$$E[\theta_i] = S_\alpha(\tau_{\text{det}}) = P(\text{Gamma}(\alpha, \beta) > \tau_{\text{det}})$$

which is **independent of** ψ.

Proof. Write $\theta_i = S_\psi(\tau_{\text{det}} - l_i) = P(\varepsilon > \tau_{\text{det}} - l_i)$ where $\varepsilon \sim \text{Gamma}(\psi\alpha, \beta)$ is independent of $l_i \sim \text{Gamma}((1-\psi)\alpha, \beta)$. Taking expectations:

$$E[\theta_i] = P(\varepsilon + l_i > \tau_{\text{det}}) = P(\text{Gamma}(\alpha, \beta) > \tau_{\text{det}})$$

by the convolution property $\text{Gamma}(\psi\alpha, \beta) + \text{Gamma}((1-\psi)\alpha, \beta) \overset{d}{=} \text{Gamma}(\alpha, \beta)$. $\square$

Interpretation. Since the population-level profile $A(\tau) = \text{Gamma}(\alpha, \beta)$ is invariant across ψ, any detection mechanism that doesn't "see" the individual's profile shape produces ψ-independent TE. All ψ-dependence must come from detection being coupled to individual biology.

How mode-anchored detection breaks invariance

For screening, $\tau_{\text{det}} \in \mathcal{W}_i = [l_i + m_\psi - d_{\text{pre}},\ l_i + m_\psi + d_{\text{post}}]$. Write $\tau_{\text{det}} = l_i + m_\psi + V$ where $V \in [-d_{\text{pre}}, d_{\text{post}}]$. Then l_i cancels:

$$\theta_i = \eta \cdot S_\psi(m_\psi + V + \delta_{\text{act}})$$

The convolution can't fire because l_i appears on both sides of the comparison $\varepsilon > \tau_{\text{det}} - l_i = m_\psi + V + \delta_{\text{act}}$. We're left comparing a $\text{Gamma}(\psi\alpha, \beta)$ variable to its own mode plus an offset — a fundamentally ψ-dependent quantity.

Monotonicity of the supra-modal mass

Proposition. Let $a = \psi\alpha$. Then $S_\psi(m_\psi) = P(\text{Gamma}(a, \beta) > \text{mode})$ is strictly decreasing from 1 to $1/2$ as a increases from 0 to ∞.

Proof sketch (integer a). By the Gamma–Poisson duality:

$$P(\text{Gamma}(a, 1) > a - 1) = P(\text{Poisson}(a - 1) \leq a - 1)$$

This is $P(\text{Poisson}(\lambda) \leq \lambda)$ with $\lambda = a - 1$, which is strictly decreasing in λ and converges to $1/2$ by the CLT. At $a \leq 1$: mode = 0, and $S_\psi(0) = 1$.

Numerical verification ($\beta = 2$):

$a = \psi\alpha$	**mode** m_ψ	$S_\psi(m_\psi)$
≤ 1	0	1.000
2	0.50	0.736
5	2.00	0.616
10	4.50	0.571
→ ∞	—	→ 0.500

Direction depends on detection offset

Define $v = V + \delta_{\text{act}}$. Then $\theta_i = \eta \cdot S_\psi(m_\psi + v)$, and:

- $v \leq 0$ (isolation at or before the mode): $S_\psi(m_\psi + v) \geq S_\psi(m_\psi)$, **decreasing in ψ** — spike profiles benefit more.
- $v > 0$ **sufficiently large** (isolation well after the mode): $S_\psi(m_\psi + v)$ is **increasing in ψ** — smooth profiles retain more tail mass and benefit more.

The crossover occurs near $v = 0$. The net screening effect averages over $V \in [-d_{\text{pre}}, d_{\text{post}}]$, so the balance of pre-mode vs post-mode detection determines which profile type benefits more overall.

> *The overlap rule: a unified TE formulation across all triggers*

The overlap rule

Unified formulation

Define the **isolation offset** $D = \tau_{\text{iso}} - l_i$ (time from biological onset to isolation) and a **transmission quantum** $\varepsilon \sim \text{Gamma}(\psi\alpha, \beta)$, drawn independently from the individual biological profile. Then:

> $$\text{TE} = \eta \cdot P(\text{det}) \cdot P(\varepsilon > D)$$

TE is the probability that a random quantum of transmission falls after isolation.

Mode-anchored triggers (screening, symptoms)

For these triggers, l_i cancels from D, giving $D = m_\psi + D_0$ where D_0 is ψ-free. Defining the **mode-centred profile** $\tilde{\varepsilon} = \varepsilon - m_\psi$:

$$\text{TE} = \eta \cdot P(\text{det}) \cdot P(\tilde{\varepsilon} > D_0)$$

Trigger	D_0 (isolation offset from mode)
Screening	$V + \delta_{\text{act}}$, where $V \in [-d_{\text{pre}}, d_{\text{post}}]$
Symptoms	$\delta_{\text{sym}} + \delta_{\text{act}}$

The pithy rule

As $\psi \to 0$, the mode-centred profile $\tilde{\varepsilon}$ concentrates at 0 (all transmission at the peak):

>

> $$\mathrm{TE}_{\mathrm{spike}} \;\approx\; \eta \cdot P(\mathrm{det}) \cdot P(D_0 < 0)$$

Spike profiles convert TE into a binary bet: was isolation before the peak? Smooth profiles give a graded overlap $E[S_\psi(m_\psi + D_0)]$ that always captures some tail.

- **Spikes win** when $P(D_0 < 0)$ is large (isolation typically before the mode).
- **Smooth wins** when $P(D_0 < 0)$ is small (isolation typically after the mode).

This explains the screening–symptom reversal in one line: screening with large d_{pre} puts D_0 mostly negative → spikes benefit. Symptoms with $\mu_{\mathrm{sym}} > 0$ put D_0 entirely positive → smooth benefits.

Contact tracing: convolution absorption

For a traced contact j, the Gamma convolution absorbs l_j:

$$E_{l_j,\varepsilon'_j}[\theta_j] = \eta \cdot P(\varepsilon'_j + l_j > \tau_{\mathrm{iso}}^{(j)}) = \eta \cdot S_\alpha(\tau_{\mathrm{iso}}^{(j)})$$

replacing S_ψ with the ψ-independent population survival function S_α. All ψ-dependence enters through the isolation timing $D^{\mathrm{CT}} = \max(0,\, m_\psi + \delta_{\mathrm{sym}}^A + \delta_{\mathrm{trace}} - \varepsilon_j) + \delta_{\mathrm{act}}$, not through the profile. Structurally:

Component	Screening / Symptoms	Contact tracing
Survival function	S_ψ (ψ-dependent)	S_α (ψ-independent)
Argument	$m_\psi + D_0$ (D_0 ψ-free)	D^{CT} (ψ-dependent)

In both cases, the Gamma convolution absorbs one ψ-dependent component, and the other remains.

Key assumptions

The overlap rule and its consequences rest on the following assumptions. These are numbered for reference; violations of each are noted.

The individual infectiousness profile is $a_i(\tau) = R_0 \cdot f_\psi(\tau - l_i)$ with $l_i \sim \mathrm{Gamma}((1-\psi)\alpha, \beta)$ and $f_\psi = \mathrm{Gamma}(\psi\alpha, \beta)$ density. This is the specific functional form that makes $l_i + \varepsilon \sim \mathrm{Gamma}(\alpha, \beta)$. Without Gamma additivity, the convolution invariance fails, and TE at fixed detection time would depend on ψ.

ψ

The half-widths d_{pre} and d_{post} do not depend on ψ. This is the most debatable assumption. In reality, spikier profiles have higher peak viral loads but shorter durations above the limit of detection, potentially narrowing the detectability window. If the window width w shrinks with ψ, this reduces $P(\mathrm{det})$ for spikes, partially or fully offsetting their advantage. The overlap rule still holds formally (with ψ-dependent w), but the clean separation $D_0 = V + \delta_{\mathrm{act}}$ with ψ-free D_0 would break.

The detectability window is centred on $l_i + m_\psi$. This is what causes l_i to cancel from the isolation offset D. If detection were anchored to a different feature — e.g., a fixed viral load threshold that doesn't track the mode, or the mean $l_i + \psi\alpha/\beta$ instead of the mode $l_i + (\psi\alpha - 1)/\beta$ — the cancellation would differ by an additive ψ-dependent shift $1/\beta$. This is a minor effect (for the mode vs. mean distinction, the difference is $1/\beta$, independent of ψ), but matters conceptually: the overlap rule requires that the anchor point shifts with l_i one-for-one.

The offset V (position of the first positive test within the window) is independent of ε (the individual's own biological jitter). This holds because the screening schedule is external to the individual's profile. Similarly, δ_{sym} is independent of ε (symptom timing doesn't depend on which specific quanta of transmission the individual produces). **This assumption does not require perfect sensitivity** ($p_{\mathrm{sens}} = 1$). With imperfect sensitivity, V is stochastically larger (failed early tests push detection later in the window), but V remains ψ-independent. The distribution of D_0 changes with p_{sens}, but the overlap rule $\mathrm{TE} = \eta \cdot P(\mathrm{det}) \cdot P(\tilde{\varepsilon} > D_0)$ holds for any p_{sens}.

$\tau_{\text{sym}} = l_i + m_\psi + \delta_{\text{sym}}$. This is the symptom-trigger analogue of A3. It means symptoms track the individual's peak infectiousness, shifted by a ψ-independent random variable δ_{sym}. If symptoms were instead anchored to infection time (i.e., $\tau_{\text{sym}} = t_i + \delta'_{\text{sym}}$), then l_i would not cancel, the convolution invariance would apply, and TE would be ψ-independent — all the interesting structure would vanish.

Result	Requires
Convolution invariance (fixed-time TE is ψ-free)	A1
Overlap rule: $\text{TE} = \eta \cdot P(\det) \cdot P(\tilde{\varepsilon} > D_0)$	A1, A3, A4
D_0 is ψ-independent (clean separation)	A1, A2, A3, A4
Symptom-trigger l_i cancellation	A1, A5
Contact tracing l_j absorption ($S_\psi \to S_\alpha$)	A1

The equivalence point of the overlap rule

> *Where exactly is the crossover — the isolation offset D_0^* at which neither spike nor smooth profiles are preferred?*

The question

The overlap rule gives the fraction of transmission averted as

$$\frac{\theta}{\eta} = S_\psi(m_\psi + D_0) = \Pr\big(\varepsilon > m_\psi + D_0\big), \qquad \varepsilon \sim \mathrm{Gamma}(\psi\alpha,\, \beta).$$

where $m_\psi = (\psi\alpha - 1)/\beta$ is the mode. Equivalently, writing $\tilde{\varepsilon} = \varepsilon - m_\psi$ for the mode-centred profile:

$$\frac{\theta}{\eta} = \Pr(\tilde{\varepsilon} > D_0).$$

The crossover D_0^* is the value at which this probability is **independent of** ψ — i.e. the isolation offset at which spikes and smooth profiles avert the same fraction.

Why $D_0 = 0$ is not the answer

At $D_0 = 0$, we evaluate $\Pr(\varepsilon > m_\psi) = \Pr\big(\mathrm{Gamma}(\psi\alpha, \beta) > \mathrm{mode}\big)$. The Gamma density is right-skewed, so **more than half the mass lies above the mode**. The degree of skewness depends on the shape parameter $a = \psi\alpha$:

ψ	$a = \psi\alpha$	$\Pr(\varepsilon > \mathrm{mode})$
0.1	1	1.000 (exponential: mode = 0)
0.3	3	0.677
0.5	5	0.629
1.0	10	0.587

At $D_0 = 0$, spike profiles still have a large advantage (S ranges from 1.0 down to 0.59). This makes sense: the exponential has all its mass to the right of its mode (which is zero), while more symmetric bell curves have progressively less.

The median–mode gap

The natural 50-50 split point of any distribution is its **median**, not its mode. If we set D_0 equal to the median–mode gap, then $\Pr(\tilde{\varepsilon} > D_0) = \Pr(\varepsilon > \mathrm{median}) = \frac{1}{2}$ regardless of shape.

For a $\mathrm{Gamma}(a, \beta)$ distribution, a classical asymptotic expansion gives:

$$\text{median} \;=\; \frac{a - \frac{1}{3} + O(a^{-1})}{\beta}$$

while the mode is $(a-1)/\beta$. The gap is therefore:

> $$D_0^* \;=\; \text{median} - \text{mode} \;\approx\; \frac{2}{3\beta}$$

with corrections of order $O(1/(a\beta)) = O(1/(\psi\alpha\beta))$. The key point: **this gap is approximately independent of the shape parameter** $a = \psi\alpha$, so it serves as an (approximate) universal crossover.

Numerical verification ($\alpha = 10, \beta = 2$)

With our standard parameters, $D_0^* = 2/(3 \cdot 2) = 1/3 \approx 0.333$.

Median–mode gap (exact, via `qgamma`):

ψ	a	**mode**	**median**	**gap**	$2/(3\beta)$
0.1	1	0.000	0.347	0.347	0.333
0.3	3	1.000	1.337	0.337	0.333
0.5	5	2.000	2.336	0.336	0.333
1.0	10	4.500	4.834	0.334	0.333

The gap is remarkably stable (0.334–0.347), confirming the $2/(3\beta)$ approximation.

Survival function at $D_0 = 1/3$:

ψ	$S_\psi(m_\psi + 1/3)$
0.1	0.513
0.3	0.502

0.5	0.501
1.0	0.500

All values cluster tightly around 0.50, confirming that at $D_0 = 1/3$ neither spike nor smooth is preferred.

Exact pairwise crossings (found numerically by bisection):

Pair	D_0^* **(exact)**	S **at crossing**
$\psi = 0.1$ vs $\psi = 1.0$	0.351	0.496
$\psi = 0.1$ vs $\psi = 0.5$	0.353	0.493
$\psi = 0.3$ vs $\psi = 1.0$	0.340	0.499
$\psi = 0.5$ vs $\psi = 1.0$	0.338	0.499

All crossings fall in the narrow band $D_0^* \in [0.338, 0.353]$, bracketing the prediction $2/(3\beta) = 0.333$.

Why the mean doesn't work

One might guess the crossover is at the **mean**–mode gap, which is exactly $1/\beta$ for all Gamma shapes. But at $D_0 = 1/\beta = 0.5$, we evaluate $\Pr(\varepsilon > \text{mean})$, and this quantity is *not* shape-independent:

ψ	$\Pr(\varepsilon > \text{mean})$
0.1	0.368 $(= e^{-1})$
0.5	0.440
1.0	0.458

The mean sits too far to the right of the mode, well into smooth-preferred territory. The mean–mode gap $1/\beta$ overestimates the crossover; the median–mode gap $2/(3\beta)$ nails it.

Summary

> The spike–smooth equivalence point is **not** at $D_0 = 0$ (the mode) but at
>
> $$D_0^* \approx \frac{2}{3\beta}$$
>
> i.e. the **median–mode gap** of the Gamma distribution. This gap is approximately independent of shape, with corrections $O(1/(\psi\alpha\beta))$.
>
> **Interpretation:** The mode splits the Gamma density asymmetrically — the right tail is always heavier. You must go slightly past the mode (to the median) before the tail fraction reaches 50% and becomes shape-independent. For our standard parameters ($\beta = 2$): $D_0^* \approx 1/3$ day. In biological terms, isolation must begin about **8 hours after the peak** for punctuation to be irrelevant.

General crossover theorem: beyond the Gamma

> *Does the median–mode crossover hold for any biological profile, not just the Gamma? If we used a Normal, would Do* = 0? If left-skewed, would Do* < 0?*

Setup

Consider *any* unimodal distribution family $\{F_\psi\}$ for the individual biological profile ε, parameterized by a concentration parameter ψ. As $\psi \to 0$, the profile concentrates into a spike at its mode m_ψ. The overlap rule gives the fraction of transmission averted as

$$\frac{\theta}{\eta} = S_\psi(m_\psi + D_0) \;=\; \Pr(\varepsilon > m_\psi + D_0), \qquad \varepsilon \sim F_\psi.$$

The crossover D_0^* is the offset where this quantity is independent of ψ.

The theorem

The argument is elementary. For each ψ:

- $S_\psi(m_\psi + D_0)$ is a continuous, strictly decreasing function of D_0, dropping from ≈ 1 to 0.
- It passes through $\frac{1}{2}$ at exactly $D_0 = \text{median}_\psi - m_\psi$.

> **Theorem.** If the median–mode gap $\Delta_\psi := \text{median}_\psi - \text{mode}_\psi$ is approximately independent of ψ, then all survival curves pass through a common point
>
> $$\left(D_0^*,\ \tfrac{1}{2}\right), \qquad D_0^* = \Delta_\psi \approx \text{const},$$
>
> and this is the universal spike–smooth crossover. The sign of D_0^* equals the sign of the skewness:
>
> - **Right-skewed** (mode < median): $D_0^* > 0$. Spikes preferred for early isolation.
> - **Symmetric** (mode = median): $D_0^* = 0$. Crossover exactly at the peak.
> - **Left-skewed** (mode > median): $D_0^* < 0$. Spikes preferred for late isolation.

If Δ_ψ depends on ψ, the pairwise crossings scatter over a range of D_0 values and no single clean crossover exists.

Why the Gamma has a clean crossover

For $\text{Gamma}(a, \beta)$, the **mean–mode gap** is

$$\text{mean} - \text{mode} = \frac{a}{\beta} - \frac{a-1}{\beta} = \frac{1}{\beta} \qquad \text{(exactly shape-independent)}.$$

Pearson's classical relation $\text{mean} - \text{mode} \approx 3(\text{mean} - \text{median})$ then gives

$$\text{median} - \text{mode} \;\approx\; \tfrac{2}{3}(\text{mean} - \text{mode}) \;=\; \frac{2}{3\beta}$$

with corrections $O(1/(a\beta))$. The leading term is independent of $a = \psi\alpha$, giving the clean crossover. Numerically:

$a = \psi\alpha$	**median – mode**	$2/(3\beta)$
1	0.347	0.333
5	0.335	0.333
10	0.334	0.333
50	0.334	0.333

Stable to three significant figures across a 50-fold range in shape.

Counterexample: the log-normal has no clean crossover

For comparison, consider a log-normal family with the same first two moments as the Gamma ($\text{mean} = a/\beta$, $\text{var} = a/\beta^2$). This requires $\sigma^2 = \log(1 + 1/a)$.

a	**median – mode**
1	0.177
5	0.380
10	0.433
50	0.485

The gap varies by nearly 3×. The pairwise crossings scatter accordingly:

Pair	D_0^*
$a = 2$ vs $a = 10$	0.188
$a = 5$ vs $a = 20$	0.312

$$a = 10 \text{ vs } a = 20 \quad 0.366$$

No single crossover exists — the "equivalence offset" depends on which pair of profiles you compare.

Left-skewed verification: reflected Gamma

A reflected Gamma ($X = C - \text{Gamma}(a, \beta)$) is left-skewed. By reflection, its median–mode gap is

$$\Delta = (C - \text{median}_{\text{Gamma}}) - (C - \text{mode}_{\text{Gamma}}) = -(\text{median}_{\text{Gamma}} - \text{mode}_{\text{Gamma}}) = -\frac{2}{3\beta}$$

Confirmed numerically: the gap is ≈ -0.334 for all a. So for left-skewed profiles, the crossover is at $D_0^* \approx -1/3$ — isolation must begin slightly *before* the peak for punctuation to be irrelevant.

Summary

> **General result:** The spike–smooth crossover occurs at $D_0^* = \text{median} - \text{mode}$ of the biological infectiousness profile. The sign tracks the skewness:
>
> $$D_0^* \begin{cases} > 0 & \text{right-skewed (e.g. Gamma)} \\ = 0 & \text{symmetric (e.g. Normal)} \\ < 0 & \text{left-skewed} \end{cases}$$
>
> **The Gamma is special:** A *clean* crossover (single D_0^* for all concentration levels) requires $\text{median} - \text{mode}$ to be approximately shape-independent. The Gamma has this property because its mean–mode gap is exactly $1/\beta$, and by Pearson's relation, the median–mode gap is $\approx 2/(3\beta)$. Other skewed families (log-normal, Weibull) lack this and have no single crossover.
>
> **Physical interpretation:** Because real infectiousness profiles are right-skewed (fast rise, slow decay), the crossover lies slightly past the peak — isolation must begin after the mode (by

> $\approx 2/(3\beta)$ days) for punctuation to be irrelevant. This is ultimately a consequence of the asymmetry between the rising and falling phases of infectiousness.

Random detection times: variance helps spikes

> *What if D_0 is random? Is the condition for spike preference simply $E[D_0] < D_0^*$?*

Setup

When detection timing is uncertain, $D_0 \sim G$ with mean μ and variance σ^2, and the expected fraction averted becomes

$$h(\psi) \;=\; E_{D_0}\big[\tilde{S}_\psi(D_0)\big] \;=\; \int \tilde{S}_\psi(d)\, g(d)\, \mathrm{d}d \;=\; \Pr(\tilde{\varepsilon} > D_0)$$

where $\tilde{\varepsilon} \perp\!\!\!\perp D_0$. Spikes are preferred when $h(\psi_{\text{spike}}) > h(\psi_{\text{smooth}})$.

Leading-order answer: yes, $E[D_0] < D_0^*$

For small variance, expand about the mean:

$$h(\psi) \;\approx\; \tilde{S}_\psi(\mu) + \tfrac{\sigma^2}{2}\, \tilde{S}''_\psi(\mu)$$

The first term gives the deterministic result: spike wins iff $\mu < D_0^*$. So to leading order, the user's intuition is correct — **the mean detection offset must fall before the median–mode gap**.

The variance correction: spikes benefit from noise

The second-order term reveals a surprise. Since $\tilde{S}'' = -\tilde{f}'$, and the density $\tilde{f}$ is decreasing past the mode (at $D_0^* > 0$), we have $\tilde{f}'(D_0^*) < 0$ and therefore $\tilde{S}''(D_0^*) > 0$ — the survival function is **convex** in this region.

By Jensen's inequality: $E[\tilde{S}(D_0)] > \tilde{S}(E[D_0])$. Both spike and smooth benefit from this convexity uplift, but the spike benefits **far more** because its survival function is steeper (more convex):

	$\tilde{S}(E[D_0])$	$E[\tilde{S}(D_0)]$	**Jensen boost**
$\psi = 0.1$ (spike)	0.513	0.551	**+0.037**
$\psi = 1.0$ (smooth)	0.500	0.511	+0.011

(At $E[D_0] = D_0^*,\ \sigma = 1,\ D_0 \sim \mathcal{N}(\mu, \sigma^2)$)

The spike's Jensen boost is 3.5× larger than the smooth's. So at the deterministic crossover $E[D_0] = D_0^*$, the spike is *still winning* when $\sigma > 0$.

The effective crossover shifts rightward

Setting $h(\psi_1) = h(\psi_2)$ and Taylor-expanding around D_0^* gives

$$D_0^{*,\text{eff}} \;\approx\; D_0^* + c\,\sigma^2, \qquad c \;=\; \frac{\tilde{f}'_{\text{smooth}}(D_0^*) - \tilde{f}'_{\text{spike}}(D_0^*)}{2\big[\tilde{f}_{\text{spike}}(D_0^*) - \tilde{f}_{\text{smooth}}(D_0^*)\big]}$$

Since the spike density is steeper ($|\tilde{f}'_{\text{spike}}| \gg |\tilde{f}'_{\text{smooth}}|$) and taller ($\tilde{f}_{\text{spike}} > \tilde{f}_{\text{smooth}}$) at D_0^*, we get $c > 0$. Numerically, $c \approx 1.3$ for our standard parameters, and:

σ	$D_0^{*,\text{eff}}$	**Shift from D_0^***
0 (deterministic)	0.333	—
0.25	0.417	+0.084
0.50	0.496	+0.162
1.00	0.604	+0.270

1.50	0.674	+0.340
2.00	0.719	+0.386

The crossover shifts *rightward* with increasing variance — **noise in detection timing favours spikes**.

Physical interpretation

> The condition for spike preference is:
>
> $$E[D_0] \lesssim \frac{2}{3\beta} + c\,\sigma_{D_0}^2$$
>
> which is *more lenient* than the deterministic condition $D_0 < 2/(3\beta)$.
>
> **Why does noise help spikes?** The spike's transmission is an all-or-nothing bet. When detection timing is noisy:
>
> - Lucky early detections ($D_0 \ll 0$) capture the spike's *entire* burst — a windfall.
> - Unlucky late detections ($D_0 \gg 0$) miss nothing extra — the spike already fired.
>
> The payoff is asymmetric: the upside of early detection (avert everything) exceeds the downside of late detection (miss everything, but you'd have missed most of it anyway). For smooth profiles, the payoff is more linear — early or late detection matters proportionally. This asymmetry is Jensen's inequality at work: the spike's step-like survival function is convex, making it the natural beneficiary of randomness.

General variance–skewness theorem

> *Does the "variance helps spikes" result hold for arbitrary unimodal profiles, not just the Gamma? Can we make a distribution-free argument?*

The argument

Yes — and the argument requires only one fact from elementary calculus: **a unimodal density increases before its mode and decreases after**.

Consider any unimodal, mode-centred profile $\tilde{\varepsilon}$ with density $\tilde{f}$ having its mode at $d = 0$. The survival function $\tilde{S}(d) = \Pr(\tilde{\varepsilon} > d)$ satisfies

$$\tilde{S}''(d) = -\tilde{f}'(d) \;=\; \begin{cases} < 0 & d < 0 \;\; \text{(concave: density still rising)} \\ > 0 & d > 0 \;\; \text{(convex: density falling)} \end{cases}$$

The inflection point is at $d = 0$ (the mode), and every unimodal distribution has this structure regardless of parametric form.

The crossover $D_0^* = \text{median} - \text{mode}$. For right-skewed profiles, $D_0^* > 0$ — the crossover lies in the **convex region**. For left-skewed, $D_0^* < 0$ — the crossover lies in the **concave region**.

For a density with scale σ_ψ (standard deviation), dimensional analysis gives

$$\tilde{f} \sim \frac{1}{\sigma_\psi}, \qquad |\tilde{f}'| \sim \frac{1}{\sigma_\psi^2}, \qquad |\tilde{S}''| = |\tilde{f}'| \sim \frac{1}{\sigma_\psi^2}$$

A more concentrated profile (smaller σ_ψ) has **greater curvature** $|\tilde{S}''|$ at any fixed point near the mode. In particular, at D_0^*:

$$|\tilde{S}''_{\text{spike}}(D_0^*)| \;\gg\; |\tilde{S}''_{\text{smooth}}(D_0^*)|$$

Verified numerically: for our Gamma parameters, $|\tilde{S}''(D_0^*)| = 2.05$ (spike) vs 0.04 (smooth) — a

50-fold ratio.

Define $\Delta(d) = \tilde{S}_{\text{spike}}(d) - \tilde{S}_{\text{smooth}}(d)$, which crosses zero at D_0^*. By Taylor expansion around D_0^*:

$$\Delta(D_0^* \pm t) \;\approx\; \frac{t^2}{2}\,\Delta''(D_0^*)$$

For D_0 centred at D_0^* with variance σ^2:

$$E\big[\Delta(D_0)\big] \;\approx\; \frac{\sigma^2}{2}\,\Delta''(D_0^*) \;=\; \frac{\sigma^2}{2}\big[\tilde{S}''_{\text{spike}}(D_0^*) - \tilde{S}''_{\text{smooth}}(D_0^*)\big]$$

Now everything clicks together:

Profile skewness	D_0^*	**Region**	$\tilde{S}''_{\text{spike}}$ **vs** $\tilde{S}''_{\text{smooth}}$	$E[\Delta]$	**Verdict**
Right-skewed	> 0	Convex	$+2.05 \gg +0.04$	> 0	**Variance helps spikes**
Left-skewed	< 0	Concave	$-2.05 \ll -0.04$	< 0	**Variance hurts spikes**
Symmetric	$= 0$	Inflection	$0 = 0$	$= 0$	**No effect**

Numerical verification

Three distribution families, each at $E[D_0] = D_0^*$, comparing h_{spike} vs h_{smooth}:

Right-skewed (Gamma), $D_0^* = +1/3$:

σ	h_{spike}	h_{smooth}	**Winner**
0	0.513	0.500	Spike (+0.013)

0.5	0.565	0.504	Spike (+0.061)
1.0	0.551	0.511	Spike (+0.040)

Left-skewed (Reflected Gamma), $D_0^* = -1/3$:

σ	h_{spike}	h_{smooth}	**Winner**
0	0.487	0.500	Smooth (+0.013)
0.5	0.435	0.496	Smooth (+0.061)
1.0	0.449	0.489	Smooth (+0.040)

Symmetric (Normal), $D_0^* = 0$:

σ	h_{spike}	h_{smooth}	**Winner**
0	0.5000	0.5000	Tie
0.5	0.5000	0.5000	Tie
1.0	0.5000	0.5000	Tie

All three cases match the theorem's predictions exactly.

The result

> **Theorem (distribution-free).** For *any* unimodal biological profile, the effect of variance in detection timing on the spike–smooth comparison is determined by the **skewness of the profile**:
>
> Writing $D_0^{*,\text{eff}}$ for the effective crossover (the value of $E[D_0]$ at which spike and smooth avert equally), the shift relative to the deterministic crossover D_0^* is
>
> $$D_0^{*,\text{eff}} = D_0^* + \delta, \qquad \delta \approx \frac{\sigma^2}{2} \cdot \frac{\tilde{S}''_{\text{spike}}(D_0^*) - \tilde{S}''_{\text{smooth}}(D_0^*)}{\tilde{f}_{\text{spike}}(D_0^*) - \tilde{f}_{\text{smooth}}(D_0^*)}$$

> and the sign of δ equals the sign of the skewness:
>
> - **Right-skewed** ($D_0^* > 0$): $\delta > 0$. Variance *helps* spikes. The crossover shifts right.
> - **Left-skewed** ($D_0^* < 0$): $\delta < 0$. Variance *hurts* spikes. The crossover shifts left.
> - **Symmetric** ($D_0^* = 0$): $\delta = 0$. Variance has no leading-order effect.
>
> The mechanism is Jensen's inequality applied to the survival function's curvature. Past the mode ($d > 0$), every unimodal $\tilde{S}$ is convex, and more concentrated profiles are *more* convex — so they benefit more from averaging over random D_0. Before the mode ($d < 0$), the inequality reverses: $\tilde{S}$ is concave, and concentration amplifies the penalty.
>
> Since real infectiousness profiles are universally right-skewed (fast viral rise, slow immune-mediated decline), the practical consequence is unambiguous: **noisy detection timing always favours punctuated transmission**.

> *Generation interval distortion under isolation: analytical derivation*

GI distortion and the tilted Euler-Lotka equation

The overlap rule gives $\mathrm{TE} = \eta \cdot P(\mathrm{det}) \cdot P(\tilde{\varepsilon} > D_0)$, which determines $R_{\mathrm{eff}} = R_0(1 - \mathrm{TE})$. But translating R_{eff} to a growth rate r via the Euler-Lotka equation requires knowledge of the **generation interval** (GI). Isolation preferentially removes late transmissions from smooth profiles, shortening the effective GI. A shorter GI inflates the growth rate for a given R_{eff}, partially undermining the intervention. Here we derive this effect analytically.

1. Setup and notation

An individual infected at time 0 has infectiousness profile decomposed as:

$$\tau = \ell + \varepsilon, \qquad \ell \sim \mathrm{Gamma}((1-\psi)\alpha,\, \beta), \quad \varepsilon \sim \mathrm{Gamma}(\psi\alpha,\, \beta)$$

where ℓ is the latent period, ε is the time from onset to a given transmission event, and $\psi \in (0, 1]$ controls punctuation. The population-level generation interval is $\tau \sim \mathrm{Gamma}(\alpha, \beta)$ with mean $T = \alpha/\beta$. We write $a = \psi\alpha$ and $b = (1-\psi)\alpha$ for brevity.

Isolation occurs at time $D = m_\psi + D_0$ after onset, where $m_\psi = \max(0, (a-1)/\beta)$ is the mode of the onset-to-transmission density and D_0 is the offset. A transmission at time ε after onset survives isolation iff $\varepsilon < D$.

2. The naive Euler-Lotka equation

Without GI distortion, one solves the **naive** Euler-Lotka equation using the natural GI density $g(\tau) = f_{\mathrm{Gamma}}(\tau;\, \alpha,\, \beta)$:

$$1 = R_{\mathrm{eff}} \int_0^\infty e^{-r\tau}\, g(\tau)\, d\tau = R_{\mathrm{eff}} \left(\frac{\beta}{\beta + r}\right)^\alpha$$

which yields the closed-form solution:

$$r_{\mathrm{naive}} = \beta \left(R_{\mathrm{eff}}^{1/\alpha} - 1\right)$$

3. Deriving the tilted Euler-Lotka equation

Under isolation, a transmission at generation time $\tau = \ell + \varepsilon$ contributes to epidemic growth only if it is **not averted**. For the fraction $1 - \eta$ of individuals who are never isolated, all transmissions survive. For the fraction η who are subject to isolation, transmission at time ε after onset survives iff $\varepsilon < D$. The Euler-Lotka equation becomes:

$$1 = R_0 \int_0^\infty e^{-r\tau}\, g^*(\tau;\, r)\, d\tau$$

where g^* is the **effective** generation interval density (accounting for both surviving and averted transmissions). We can decompose:

$$1 = R_0 \int_0^\infty \int_0^\infty e^{-r(\ell+\varepsilon)}\, f_\ell(\ell)\, [(1-\eta) f_\varepsilon(\varepsilon) + \eta\, f_\varepsilon(\varepsilon)\, \mathbf{1}(\varepsilon < D)]\, d\varepsilon\, d\ell$$

The ℓ-integral factors out since ℓ and ε are independent:

$$\int_0^\infty e^{-r\ell} f_\ell(\ell)\, d\ell = \left(\frac{\beta}{\beta + r}\right)^b$$

where $b = (1-\psi)\alpha$. The ε-integral splits into two terms:

$$(1-\eta) \int_0^\infty e^{-r\varepsilon} f_\varepsilon(\varepsilon)\, d\varepsilon + \eta \int_0^D e^{-r\varepsilon} f_\varepsilon(\varepsilon)\, d\varepsilon$$

The first integral is $\left(\frac{\beta}{\beta+r}\right)^a$. For the second, we use the identity for the truncated Laplace transform of a Gamma:

$$\int_0^D e^{-r\varepsilon} f_{\text{Gamma}}(\varepsilon;\, a,\, \beta)\, d\varepsilon = \left(\frac{\beta}{\beta + r}\right)^a F_{\text{Gamma}}(D;\, a,\, \beta + r)$$

where $F_{\text{Gamma}}(D; a, \beta + r) = \texttt{pgamma}(D, a, \beta + r)$ is the CDF of a $\text{Gamma}(a, \beta + r)$. This is the key "exponential tilting" identity: multiplying a Gamma density by $e^{-r\varepsilon}$ and renormalising gives a Gamma with rate $\beta + r$.

Combining everything:

$$1 = R_0 \left(\frac{\beta}{\beta + r}\right)^\alpha [(1-\eta) + \eta\, F_{\text{Gamma}}(D;\, a,\, \beta + r)]$$

Or equivalently, using the survival function $\bar{F} = 1 - F$:

> $$1 = R_0 \left(\frac{\beta}{\beta + r}\right)^\alpha \left[1 - \eta\, \bar{F}_{\text{Gamma}}(D;\, a,\, \beta + r)\right]$$

where $\bar{F}_{\text{Gamma}}(D; a, \beta + r) = \texttt{pgamma}(D, a, \beta + r, \text{lower} = \text{FALSE})$. This is the **tilted Euler-Lotka equation**.

4. The exponential tilting mechanism

Comparing the tilted equation with the naive one reveals the key insight. The naive equation uses $R_{\text{eff}} = R_0[1 - \eta\, \bar{F}_{\text{Gamma}}(D;\ a,\ \beta)]$ and the natural GI. The tilted equation evaluates the survival function at rate $\beta + r$ instead of β.

For $r > 0$ (growing epidemic), $\beta + r > \beta$, so the tilted Gamma has a **shorter mean** ($a/(\beta + r) < a/\beta$). This means more of its mass is concentrated below D, so $\bar{F}(D; a, \beta + r) < \bar{F}(D; a, \beta)$. In words: **from the perspective of the growth rate equation, fewer transmissions appear to be averted** because the exponential discounting $e^{-r\tau}$ already downweights late transmissions. The isolation truncation at D is thus partially redundant with the natural discounting.

This means $r_{\text{actual}} > r_{\text{naive}}$: the epidemic grows faster than the naive prediction from R_{eff} alone.

5. Distorted generation interval density

The effective GI density among surviving transmissions is:

$$w^*(\tau) = \frac{(1-\eta)g(\tau) + \eta \int_0^{\min(\tau,D)} f_\ell(\tau - \varepsilon) f_\varepsilon(\varepsilon)\, d\varepsilon}{1 - \text{TE}}$$

For the non-isolated fraction $(1 - \eta)$, the GI is unchanged. For the isolated fraction, only transmissions with $\varepsilon < D$ survive, giving a truncated convolution of f_ℓ and the truncated f_ε. The denominator $1 - \text{TE} = 1 - \eta\, \bar{F}_{\text{Gamma}}(D; a, \beta)$ normalises to a proper density.

6. Mean generation interval under isolation

For the isolated fraction, the mean generation time of surviving transmissions is:

$$E[\tau^* \mid \text{isolated, surviving}] = E[\ell] + E[\varepsilon \mid \varepsilon < D]$$

$$= \frac{b}{\beta} + \frac{\int_0^D \varepsilon\, f_{\text{Gamma}}(\varepsilon;\ a,\ \beta)\, d\varepsilon}{F_{\text{Gamma}}(D;\ a,\ \beta)}$$

The truncated mean of ε uses the identity:

$$\int_0^D \varepsilon\, f_{\text{Gamma}}(\varepsilon;\, a,\, \beta)\, d\varepsilon = \frac{a}{\beta}\, F_{\text{Gamma}}(D;\, a+1,\, \beta)$$

so:

$$E[\varepsilon \mid \varepsilon < D] = \frac{a}{\beta} \cdot \frac{F_{\text{Gamma}}(D;\, a+1,\, \beta)}{F_{\text{Gamma}}(D;\, a,\, \beta)}$$

The overall mean GI under isolation, mixing isolated and non-isolated individuals weighted by their transmission contributions:

$$E[\tau^*] = \frac{(1-\eta)\cdot \frac{\alpha}{\beta} + \eta \cdot F_{\text{Gamma}}(D; a, \beta) \cdot \left[\frac{b}{\beta} + \frac{a}{\beta} \cdot \frac{F_{\text{Gamma}}(D; a+1, \beta)}{F_{\text{Gamma}}(D; a, \beta)}\right]}{1 - \eta\, \bar{F}_{\text{Gamma}}(D; a, \beta)}$$

Simplifying:

> $$E[\tau^*] = \frac{(1-\eta)\cdot \frac{\alpha}{\beta} + \eta \left[\frac{b}{\beta} \cdot F_{\text{Gamma}}(D; a, \beta) + \frac{a}{\beta} \cdot F_{\text{Gamma}}(D; a+1, \beta)\right]}{1 - \eta\, \bar{F}_{\text{Gamma}}(D; a, \beta)}$$

7. Limiting cases

$\psi \to 0$

When $\psi \to 0$, $a = \psi\alpha \to 0$ and $b \to \alpha$. The individual infectious period $\varepsilon \sim \text{Gamma}(a, \beta)$ degenerates to a point mass at 0. Isolation cannot truncate a zero-duration event: $\bar{F}_{\text{Gamma}}(D; 0, \beta + r) = 0$ for $D > 0$. The tilted Euler-Lotka reduces to the naive equation. **No GI distortion for spikes.**

$\psi = 1$

When $\psi = 1$, $a = \alpha$ and $b = 0$. The entire generation interval is the infectious period ε; there is

no latent period. The tilting effect is maximal: $\bar{F}_{\text{Gamma}}(D; \alpha, \beta + r)$ can differ substantially from $\bar{F}_{\text{Gamma}}(D; \alpha, \beta)$. **Maximal GI distortion for smooth profiles.**

$\eta = 0$

With $\eta = 0$, the bracketed term in the tilted Euler-Lotka is 1, recovering the standard equation $1 = R_0(\beta/(\beta + r))^\alpha$. The naive and actual growth rates coincide. **No distortion without intervention.**

$D \to \infty$

As $D \to \infty$, $\bar{F}_{\text{Gamma}}(D; a, \beta + r) \to 0$, so no transmissions are averted. The tilted Euler-Lotka again reduces to the standard equation.

$D \to 0$

As $D \to 0$, $\bar{F}_{\text{Gamma}}(D; a, \beta + r) \to 1$ (for $a > 0$), so $1 = R_0(\beta/(\beta + r))^\alpha[1 - \eta]$, i.e. $R_{\text{eff}} = R_0(1 - \eta)$ and the GI is unchanged (all post-onset transmissions are removed equally in the tilted and untilted versions). In this case, naive and actual agree because truncation at the onset is so aggressive that the GI shape doesn't matter.

8. Growth rate bias

Define:

- $r_{\text{naive}} = \beta(R_{\text{eff}}^{1/\alpha} - 1)$ from the standard Euler-Lotka with the natural GI
- r_{actual} = root of the tilted Euler-Lotka: $R_0(\beta/(\beta + r))^\alpha[1 - \eta\,\bar{F}(D; a, \beta + r)] = 1$

The **growth rate bias** is:

$$\Delta r = r_{\text{actual}} - r_{\text{naive}} \geq 0$$

with equality when $\psi = 0$, $\eta = 0$, or $D \to \infty$. The bias is largest when ψ is large (smooth profiles), η is large (high intervention coverage), and D is moderate (truncation in the body of the distribution).

For the relative bias, consider:

$$\text{Relative bias} = \frac{r_{\text{actual}} - r_{\text{naive}}}{r_{\text{naive}}} \times 100\%$$

This quantifies the percentage overestimate of intervention effectiveness when GI distortion is ignored.

Detection Survival and Punctuation: General Theory

2026-03-23

> *Setup: $S(\tau)$ is a detection survival function (continuous, $S(0) = 1$, monotonically decreasing) and $\{a_\psi(\tau)\}$ is a family of infectiousness profiles indexed by a punctuation parameter ψ, all unimodal and integrating to R_0. More punctuated profiles are narrower/peakier. We want to understand the sign of*
>
> $$\Delta = \int_0^\infty S(\tau)\left[a_{\psi_1}(\tau) - a_{\psi_2}(\tau)\right] d\tau$$
>
> *where ψ_1 is more punctuated than ψ_2.*

1. The escaped-transmission functional

The quantity $\int S(\tau)\, a_\psi(\tau)\, d\tau$ represents the **expected infectiousness that escapes detection**. Without detection, total infectiousness is $R_0 = \int a_\psi\, d\tau$. With detection, it is reduced to $\int S \cdot a_\psi$. Defining $f_\psi = a_\psi / R_0$ as the normalised density of transmission timing, the escaped fraction is:

$$\frac{1}{R_0} \int S(\tau)\, a_\psi(\tau)\, d\tau = \mathbb{E}_{f_\psi}[S(\tau)]$$

So the question reduces to: **how does the expected value of a decreasing function S change as we concentrate the distribution f_ψ?**

2. Proposition (Single-crossing lemma)

Before considering mean-preserving contractions, there is a simpler result for the case where the difference $\delta(\tau) = a_{\psi_1}(\tau) - a_{\psi_2}(\tau)$ crosses zero exactly once.

> **Lemma (Chebyshev-type).** Suppose $\delta(\tau)$ changes sign from $+$ to $-$ at a unique crossing point τ^*, and $\int \delta \, d\tau = 0$. If S is decreasing, then
>
> $$\int_0^\infty S(\tau)\,\delta(\tau)\,d\tau \geq 0.$$

Proof. On $[0, \tau^*]$, we have $\delta(\tau) \geq 0$ and $S(\tau) \geq S(\tau^*)$. On $[\tau^*, \infty)$, we have $\delta(\tau) \leq 0$ and $S(\tau) \leq S(\tau^*)$. Therefore:

$$\int_0^\infty S\,\delta\,d\tau \geq S(\tau^*)\int_0^{\tau^*} \delta\,d\tau + S(\tau^*)\int_{\tau^*}^\infty \delta\,d\tau = S(\tau^*)\underbrace{\int_0^\infty \delta\,d\tau}_{=0} = 0. \quad \square$$

Interpretation: If profile ψ_1 has more early mass and less late mass than ψ_2, with a single crossing, then ψ_1 always has *more* transmission escaping detection. Monotone decreasing S always weights early mass more heavily. This is a **first-order** (location) effect.

But for comparing profiles that differ primarily in *spread* (not location), the difference δ typically has **two** sign changes: negative in the left tail, positive near the mode, negative in the right tail (or the reverse). This is where Jensen's inequality enters.

3. Proposition (Jensen / convex-order effect)

> **Proposition.** Suppose f_{ψ_1} and f_{ψ_2} have the same mean μ, but f_{ψ_1} is a **mean-preserving contraction** of f_{ψ_2} (i.e., $f_{\psi_1} \leq_{\text{cx}} f_{\psi_2}$ in convex order, meaning $\text{Var}_{\psi_1} < \text{Var}_{\psi_2}$). Then:

> - S **convex** on the support $\Longrightarrow \mathbb{E}_{\psi_1}[S] \leq \mathbb{E}_{\psi_2}[S]$ — **punctuation reduces escape, helps detection.**
> - S **concave** on the support $\Longrightarrow \mathbb{E}_{\psi_1}[S] \geq \mathbb{E}_{\psi_2}[S]$ — **punctuation increases escape, hurts detection.**
> - S **linear** on the support $\Longrightarrow \mathbb{E}_{\psi_1}[S] = \mathbb{E}_{\psi_2}[S]$ — **punctuation irrelevant.**

This follows directly from Jensen's inequality. To leading order in the variance difference:

$$\mathbb{E}_{\psi_1}[S] - \mathbb{E}_{\psi_2}[S] \;\approx\; \tfrac{1}{2}\, S''(\mu)\big(\mathrm{Var}_{\psi_1} - \mathrm{Var}_{\psi_2}\big)$$

Since $\mathrm{Var}_{\psi_1} < \mathrm{Var}_{\psi_2}$ (more punctuated = less spread), the sign of Δ is determined by $S''(\mu)$:

$$\boxed{\mathrm{sign}(\Delta) = -\mathrm{sign}\big(S''(\mu)\big)}$$

(where $\Delta > 0$ means the more punctuated profile has more escape).

4. Interpretation via the detection hazard

Writing $S(\tau) = \exp\big(-\int_0^\tau h(u)\,du\big)$ where $h(\tau) = -S'(\tau)/S(\tau) \geq 0$ is the detection hazard, the second derivative of S is:

$$S''(\tau) = \big[h(\tau)^2 - h'(\tau)\big]\, S(\tau)$$

Since $S > 0$, the curvature condition becomes:

$$S \text{ convex at } \tau \iff h'(\tau) < h(\tau)^2 \qquad\qquad S \text{ concave at } \tau \iff h'(\tau) > h(\tau)^2$$

In words:

- S **is convex** when the hazard is constant or slowly increasing (detection is "steady-state"). The paradigm case is exponential S (constant hazard).
- S **is concave** when the hazard is rapidly increasing (detection is "ramping up"). This happens during the onset of a detection regime — e.g., the period where symptom awareness is kicking in, or where test sensitivity is climbing.

> **Rule of thumb:** Punctuation helps detection when the detection hazard is *mature* (steady or slowly growing), and hurts detection when the hazard is *ramping up* (rapidly accelerating). The key quantity is whether $h'/h^2 \lesseqgtr 1$ at the infectiousness peak.

5. Examples

$$S(\tau) = e^{-h\tau}$$

$S'' = h^2 e^{-h\tau} > 0$ everywhere (convex). So **punctuation always helps detection**. Intuitively: a constant hazard means that each moment of delay costs the same proportional detection rate. Concentrating infectiousness in a short window gives that infectiousness less total "exposure" to the cumulative hazard.

Wait — that intuition may seem backwards; let's be precise. With exponential S and a symmetric profile centred at μ:

$$\mathbb{E}[e^{-h\tau}] = e^{-h\mu}\, M_f(h)$$

where $M_f(h) = \mathbb{E}[e^{-h(\tau-\mu)}]$ is the centred moment generating function. Since e^{-hx} is convex, $M_f(h) \geq 1$, with equality iff f is degenerate. More spread $\Rightarrow$ higher $M_f(h) \Rightarrow$ *more* escape. So the more punctuated (less spread) profile has *less* escape — punctuation helps detection. ✔

$$S(\tau) = \mathbf{1}[\tau < T]$$

S is flat (linear, with $S'' = 0$) everywhere except at $\tau = T$, where it has a concave kink. The integral is simply:

$$\mathbb{E}_f[S] = F_f(T) = P(\tau < T)$$

For mean-preserving contractions: concentrating the profile around μ pulls mass away from the tails. If $T > \mu$, there is more right-tail mass that gets pulled in below T, so $F(T)$ increases — **punctuation increases escape** (hurts detection). If $T < \mu$, mass gets pulled away from the left tail below T, so $F(T)$ decreases — **punctuation decreases escape** (helps detection).

In short: a step-function S favours the profile whose mass is most concentrated in the pre-detection window. If detection comes after the peak, punctuation is bad for detection (the peak is fully "inside" the window). If detection comes before the peak, punctuation is good (less tail leaks into the early window).

$$S(\tau) = e^{-(\tau/\lambda)^k}$$

The curvature transition occurs where $S'' = 0$, which gives:

$$\tau^* = \lambda \left(\frac{k-1}{k} \right)^{1/k} \qquad (k > 1)$$

For $\tau < \tau^*$: S is concave (ramping-up phase). For $\tau > \tau^*$: S is convex (mature phase). So:

- Infectiousness peak in the **early** (concave) region: punctuation **hurts** detection.
- Infectiousness peak in the **late** (convex) region: punctuation **helps** detection.

For $k = 2$ (Rayleigh-type survival): $\tau^* = \lambda/\sqrt{2}$. For $k \leq 1$, S is convex everywhere — same qualitative behaviour as exponential.

6. The general case: mean shift + spread change

In many parameterisations of punctuation (including the shifted-Gamma model), changing punctuation shifts both the mean and the spread of the individual profile. If profile ψ_1 has mean μ_1 and variance σ_1^2, and profile ψ_2 has mean μ_2 and variance σ_2^2, then to second order:

$$\Delta \approx R_0 \Big[S'(\bar{\mu}) \underbrace{(\mu_1 - \mu_2)}_{\text{location shift}} + \tfrac{1}{2} S''(\bar{\mu}) \underbrace{(\sigma_1^2 - \sigma_2^2)}_{\text{spread change}} \Big]$$

where $\bar{\mu} = (\mu_1 + \mu_2)/2$. The two effects can reinforce or oppose each other. In the shifted-Gamma model, for a given individual with onset l, the individual profile $f_\psi(\cdot - l)$ has mean $l + \psi\alpha/\beta$ and variance $\psi\alpha/\beta^2$, so more punctuation (lower ψ) both **shifts the mean earlier** (towards higher S, increasing escape) and **reduces variance** (effect depends on S'').

7. Critical caveat: population-level invariance for exogenous S

This is crucial for connecting to the punctuation model. In the shifted-Gamma framework, each individual has profile $a_i(\tau) = R_0\, f_\psi(\tau - l_i)$, where $l_i \sim \mathrm{Gamma}((1-\psi)\alpha, \beta)$ and $f_\psi = \mathrm{Gamma}(\psi\alpha, \beta)$. The population-level profile is:

$$A(\tau) = \mathbb{E}_l\left[R_0\, f_\psi(\tau - l)\right] = R_0 \cdot \mathrm{Gamma}(\alpha, \beta)$$

which is ψ**-invariant** (by the Gamma convolution property). Therefore, if $S(\tau)$ depends only on time-since-infection and is the **same for all individuals**, then:

$$\mathbb{E}_l\left[\int S(\tau)\, a_i(\tau; l)\, d\tau\right] = \int S(\tau)\, A(\tau)\, d\tau$$

> **Invariance result:** If $S(\tau)$ is exogenous (not coupled to individual biology), the population-level escaped transmission $\int S\, A\, d\tau$ does not depend on ψ. The individual-level Jensen effects (Section 3) average out perfectly over the distribution of onset times l_i.

This means that for the punctuation parameter ψ to matter at the population level, we need S **to be endogenous** — coupled to the individual's infectiousness profile. Natural mechanisms include:

1. **Symptom-triggered detection:** Symptom onset correlates with the infectiousness peak. For a more punctuated individual, symptom onset is more tightly linked to the (narrower) infectious period, creating a different $S_i(\tau)$ than for a smooth individual.
2. **Viral-load-triggered detection:** Test sensitivity depends on viral load, which tracks infectiousness. The effective $S_i(\tau)$ depends on the shape of $a_i(\tau)$ itself.
3. **Contact-trace-triggered detection:** Detection of individual i depends on when their infector was detected, which depends on the infector's profile.

In all these cases, S_i and a_i are positively correlated (high infectiousness $\Leftrightarrow$ high detectability), which tends to make detection more effective against smooth profiles (whose detectable window overlaps more with their infectious window).

8. Summary table

Detection regime	$S(\tau)$ curvature	Hazard condition	Punctuation effect on escape
Constant hazard (exponential)	Convex	$h' = 0$	Decreases escape (helps detection)
Slowly increasing hazard	Convex	$h' < h^2$	Decreases escape (helps detection)
Rapidly increasing hazard	Concave	$h' > h^2$	Increases escape (hurts detection)
Linear S	Zero	—	No effect (to second order)
Step function at T	Concave kink at T	—	Depends on T vs μ

The overarching message: **the curvature of the detection survival function at the infectiousness peak determines whether punctuation helps or hurts detection.** But at the population level in the shifted-profile model, exogenous S produces no net effect — the individual-level effects integrate to zero. To get a population-level effect, detection must couple to individual biology.

Mode-Anchored Detection: Breaking the Invariance

2026-03-23

> *We now consider the case where the detection survival function is* ***anchored to the mode*** *of each individual's infectiousness profile. That is, each individual i has survival function $S_i(\tau) = \tilde{S}(\tau - \tau_{\text{mode},i})$, where $\tau_{\text{mode},i}$ is the time at which individual i's infectiousness*

> *peaks and $\tilde{S}$ is a canonical survival function defined relative to the mode.*

9. The mode-anchoring transformation

In the shifted-Gamma framework, individual i has infectiousness profile $a_i(\tau) = R_0\, f_\psi(\tau - l_i)$, where $f_\psi = \mathrm{Gamma}(\psi\alpha, \beta)$ has mode at $m_\psi = (\psi\alpha - 1)/\beta$. The individual's peak infectiousness occurs at time:

$$\tau_{\mathrm{mode},i} = l_i + m_\psi$$

With mode-anchored detection, individual i's survival function is $S_i(\tau) = \tilde{S}(\tau - l_i - m_\psi)$. Substituting into the escaped-transmission integral:

$$\int_0^\infty S_i(\tau)\, a_i(\tau)\, d\tau = R_0 \int_0^\infty \tilde{S}(\tau - l_i - m_\psi)\, f_\psi(\tau - l_i)\, d\tau$$

Setting $u = \tau - l_i$:

$$= R_0 \int_0^\infty \tilde{S}(u - m_\psi)\, f_\psi(u)\, du$$

> **Key cancellation:** The onset shift l_i has dropped out entirely. The escaped transmission depends only on ψ (which determines both the profile shape f_ψ and the mode m_ψ), not on the individual's random onset time. This quantity is **identical for all individuals** sharing the same ψ, and therefore the population average equals the individual-level quantity. The invariance is broken.

10. The mode-centred residual

Define the **mode-centred residual** $\tilde{\varepsilon} = u - m_\psi$, representing the timing of a transmission event relative to the individual's infectiousness peak. Then:

$$\frac{1}{R_0} \int S_i(\tau)\, a_i(\tau)\, d\tau = \mathbb{E}_{\tilde{\varepsilon} \sim \tilde{f}_\psi}\big[\tilde{S}(\tilde{\varepsilon})\big]$$

where $\tilde{f}_\psi$ is the density of $\tilde{\varepsilon}$. The properties of $\tilde{\varepsilon}$ are:

- **Mode:** $\tilde{\varepsilon}$ has its mode at 0 by construction.
- **Support:** $\tilde{\varepsilon} \in [-m_\psi, \infty)$. For $\psi\alpha > 1$, there is mass to the left of the mode; for $\psi\alpha \leq 1$, the mode is at the left boundary and $\tilde{\varepsilon} \geq 0$.
- **Mean:** $\mathbb{E}[\tilde{\varepsilon}] = \underbrace{\psi\alpha/\beta}_{\text{mean of } f_\psi} - \underbrace{(\psi\alpha - 1)/\beta}_{m_\psi} = 1/\beta$, which is **constant across** ψ.
- **Variance:** $\mathrm{Var}(\tilde{\varepsilon}) = \psi\alpha/\beta^2$, which **decreases linearly in** ψ (more punctuated $\to$ less variance).
- **Skewness:** $\mathrm{skew}(\tilde{\varepsilon}) = 2/\sqrt{\psi\alpha}$, which increases with punctuation.

The constancy of $\mathbb{E}[\tilde{\varepsilon}] = 1/\beta$ is important: it means that as ψ varies, the mode-centred profile undergoes a **mean-preserving contraction**. The mean of a Gamma always exceeds its mode by exactly $1/\beta$, regardless of the shape parameter. So the only effect of changing ψ (to leading order) is a change in variance.

11. Main result: curvature of $\tilde{S}$ at the mode

Since $\tilde{\varepsilon}$ undergoes a mean-preserving contraction as ψ decreases, the Jensen result from Section 3 applies directly. Writing the escaped fraction as a Taylor expansion around the mean $\mu_{\tilde{\varepsilon}} = 1/\beta$:

$$\mathbb{E}\big[\tilde{S}(\tilde{\varepsilon})\big] = \tilde{S}(1/\beta) + \tfrac{1}{2}\,\tilde{S}''(1/\beta)\,\mathrm{Var}(\tilde{\varepsilon}) + O\big(\mathrm{Kurt}\cdot\mathrm{Var}^2\big)$$

Since $\mathrm{Var}(\tilde{\varepsilon}) = \psi\alpha/\beta^2$:

$$\boxed{\frac{\partial}{\partial\psi}\,\mathbb{E}\big[\tilde{S}(\tilde{\varepsilon})\big] \;\approx\; \frac{\alpha}{2\beta^2}\,\tilde{S}''\!\left(\frac{1}{\beta}\right)}$$

> **Direction rule:** The sign of the ψ-effect on escaped transmission is determined by the curvature of $\tilde{S}$ evaluated at $1/\beta$ — the **mean–mode gap** of the individual profile. Specifically:
>
> - $\tilde{S}''(1/\beta) > 0$ (convex): increasing ψ (less punctuated, more spread) **increases** escape. Equivalently, **punctuation helps detection**.

> - $\tilde{S}''(1/\beta) < 0$ (concave): increasing ψ **decreases** escape. **Punctuation hurts detection.**
> - $\tilde{S}''(1/\beta) = 0$ (linear): no second-order effect of punctuation.
>
> Note: this is the curvature of $\tilde{S}$ at the *mean* of $\tilde{\varepsilon}$ (i.e., at one mean–mode gap to the right of the mode), not at the mode itself.

12. Why $1/\beta$, not 0?

It may be surprising that the relevant evaluation point is $\tilde{S}(1/\beta)$ rather than $\tilde{S}(0)$. The reason is that the Jensen expansion is around the *mean* of the distribution, not its mode. For a right-skewed distribution (Gamma), the mean lies to the right of the mode by exactly $1/\beta$. So the relevant curvature is the curvature of $\tilde{S}$ at the point "one mean–mode gap to the right of peak infectiousness."

In our parameterisation with $\beta = \alpha/T$ (where T is the mean generation interval), the mean–mode gap is:

$$\frac{1}{\beta} = \frac{T}{\alpha}$$

With $T = 5$ days and $\alpha = 10$: the gap is 0.5 days. So we evaluate $\tilde{S}''$ half a day after peak infectiousness. As $\alpha \to \infty$ (the GI distribution sharpens), this gap shrinks to zero and we recover evaluation at the mode.

13. Geometric interpretation: supra-modal mass

The supra-modal mass $P(\tilde{\varepsilon} > 0) = P(\mathrm{Gamma}(\psi\alpha, \beta) > m_\psi)$ gives a concrete geometric quantity related to the step-function limit of $\tilde{S}$. Recall from earlier work:

$\psi\alpha$	**Supra-modal mass**	**Infra-modal mass**
≤ 1	1.000	0.000
2	0.736	0.264

5	0.616	0.384
10	0.571	0.429
$\to \infty$	0.500	0.500

For a step-function $\tilde{S}$ that drops from 1 to 0 at $\tilde{\varepsilon} = 0$ (i.e., instantaneous detection at the mode), the escaped transmission equals the **infra-modal mass**. This is strictly increasing in $\psi\alpha$, confirming that **more punctuated profiles escape less** under mode-triggered detection. The step function has a concave kink at 0, but since all the mass to the left of 0 survives and all the mass to the right doesn't, what matters is how much mass is packed before vs after the mode — and spike profiles pack almost everything after.

But notice: the *prevented* transmission (supra-modal mass) is large for *all* values of $\psi\alpha$ (always above 50%). This means mode-anchored detection is inherently effective regardless of punctuation. The question is only about the *marginal* effect of punctuation.

14. The general framework (beyond shifted-Gamma)

The results above rely on two properties of the Gamma distribution:

1. The mean–mode gap $1/\beta$ is constant across shape parameters.
2. The family is ordered by convex order (mean-preserving contraction as shape decreases).

For a **general unimodal family** f_ψ with mode m_ψ and mean μ_ψ, the escaped transmission under mode-anchored $\tilde{S}$ is:

$$\mathbb{E}[\tilde{S}(\tilde{\varepsilon}_\psi)] \;\approx\; \tilde{S}(\mu_\psi - m_\psi) + \tfrac{1}{2}\,\tilde{S}''(\mu_\psi - m_\psi)\,\mathrm{Var}_\psi(\tilde{\varepsilon}) + \cdots$$

If both the mean–mode gap $g_\psi = \mu_\psi - m_\psi$ and the variance change with ψ, we have two effects:

$$\frac{d}{d\psi}\,\mathbb{E}[\tilde{S}(\tilde{\varepsilon}_\psi)] \;\approx\; \tilde{S}'(g_\psi)\,\frac{dg_\psi}{d\psi} + \tfrac{1}{2}\,\tilde{S}''(g_\psi)\,\frac{d\mathrm{Var}_\psi}{d\psi}$$

The first term is a **location effect**: if the mean–mode gap changes with punctuation, $\tilde{S}$ is evaluated at a different point, creating a first-order effect. The second term is the familiar

curvature/Jensen effect. For the Gamma family, $dg/d\psi = 0$, so the location effect vanishes and the curvature effect is the whole story.

15. Connection to the detection hazard

Writing $\tilde{S}(t) = \exp\!\left(-\int_0^t \tilde{h}(u)\,du\right)$, where now $\tilde{h}(t)$ is the detection hazard as a function of time-since-mode:

$$\tilde{S}''(t) = \left[\tilde{h}(t)^2 - \tilde{h}'(t)\right]\tilde{S}(t)$$

So the curvature condition at the evaluation point $t_0 = 1/\beta$ is:

$$\tilde{S}''(t_0) > 0 \iff \tilde{h}'(t_0) < \tilde{h}(t_0)^2$$

In words: **punctuation helps detection when the detection hazard is "mature" (not accelerating too fast) one mean–mode gap after the infectiousness peak.**

For the specific mechanisms the user has in mind:

Suppose an individual is detectable in the window $[m - d_l,\ m + d_r]$ (i.e., $\tilde{\varepsilon} \in [-d_l, d_r]$), and tests arrive as a Poisson process with rate λ. Then:

$$\tilde{S}(t) = \begin{cases} 1 & t < -d_l \\ e^{-\lambda(t+d_l)} & -d_l \le t \le d_r \\ e^{-\lambda(d_l+d_r)} & t > d_r \end{cases}$$

Within the detectability window, $\tilde{S}$ is exponential — **convex**. So $\tilde{S}''(1/\beta) > 0$ whenever $1/\beta$ falls within $[-d_l, d_r]$, and **punctuation helps detection**.

If instead tests arrive at a regular cadence with random phase (uniform on $[0, c]$), then within the window $\tilde{S}(t) \approx \max(0, 1 - (t + d_l)/c)$, which is **linear**: $\tilde{S}'' = 0$. In this case, **the second-order punctuation effect vanishes**, and one must look to higher-order terms (skewness).

If symptoms appear at time $m + \delta$ where δ has density $p(\delta)$, and isolation occurs immediately upon symptoms, then:

$$\tilde{S}(t) = P(\delta > t) = 1 - F_\delta(t)$$

so $\tilde{S}'(t) = -p(t)$ and $\tilde{S}''(t) = -p'(t)$. Therefore:

$$\tilde{S}''(1/\beta) > 0 \iff p'(1/\beta) < 0$$

That is, **punctuation helps detection iff the symptom-onset density is decreasing** at one mean–mode gap after the infectiousness peak. If the symptom density is still ramping up at that point (many people become symptomatic shortly after peak infectiousness), punctuation hurts. If symptoms are most likely at or before the peak and the density is declining by $1/\beta$, punctuation helps.

16. Summary

> **Mode-anchored detection breaks the ψ-invariance** because the onset shift l_i cancels between the detection and infectiousness processes. The escaped transmission becomes:
>
> $$R_{\text{eff}} = R_0 \cdot \mathbb{E}_{\tilde{\varepsilon} \sim \tilde{f}_\psi}\left[\tilde{S}(\tilde{\varepsilon})\right]$$
>
> where $\tilde{\varepsilon}$ is the mode-centred residual (mean $1/\beta$, variance $\psi\alpha/\beta^2$). The ψ-sensitivity is:
>
> $$\frac{\partial R_{\text{eff}}}{\partial \psi} \approx \frac{R_0\,\alpha}{2\beta^2}\ \tilde{S}''\!\left(\frac{1}{\beta}\right)$$
>
> Punctuation helps detection when $\tilde{S}$ is **convex** at $1/\beta$ (detection hazard mature / slowly growing), and hurts when $\tilde{S}$ is **concave** (hazard ramping up). The evaluation point $1/\beta = T/\alpha$ is the mean–mode gap of the Gamma distribution — typically a fraction of a day for realistic parameters.

Connecting the median–mode and mean–mode results

2026-03-23

> *The earlier analysis (Sections on "The equivalence point of the overlap rule") found that the spike–smooth crossover for step-function detection occurs at $D_0^* = \text{median} - \text{mode} \approx 2/(3\beta)$. The current analysis (Section 11) finds that the ψ-sensitivity of escaped transmission under general $\tilde{S}$ depends on $\tilde{S}''(1/\beta)$ — the curvature at the mean–mode gap. How do these relate?*

17. The two reference points

Both results identify a special point in mode-relative time where punctuation "doesn't matter." But they do so at different levels of generality:

	Earlier result	**Current result**
Detection model	Step function at D_0: instant isolation at a fixed offset from the mode	General smooth $\tilde{S}(\tilde{\varepsilon})$
Question asked	At what D_0 is escaped transmission ψ-independent?	In which direction does ψ push escaped transmission?
Answer	$D_0^* = \text{median} - \text{mode} \approx 2/(3\beta)$	$\text{sign}(\partial R_{\text{eff}}/\partial\psi) = \text{sign}(\tilde{S}''(1/\beta))$
Reference point	Median of $\tilde{\varepsilon}$ $(\approx 2/(3\beta))$	Mean of $\tilde{\varepsilon}$ $(= 1/\beta)$
Exactness	Exact (for step-function $\tilde{S}$)	Second-order approximation (for smooth $\tilde{S}$)

18. Why the median is exact for step functions

For a step function $\tilde{S}(t) = \mathbf{1}[t < D_0]$, the escaped transmission is simply the CDF:

$$\mathbb{E}[\tilde{S}(\tilde{\varepsilon})] = P(\tilde{\varepsilon} < D_0) = \tilde{F}_\psi(D_0)$$

The crossover requires $\tilde{F}_\psi(D_0)$ to be independent of ψ. By definition, **the median is the unique point where $\tilde{F}_\psi = \frac{1}{2}$** for any distribution. So if the median–mode gap is approximately ψ-independent (as it is for the Gamma), then $\tilde{F}_\psi(D_0^*) \approx \frac{1}{2}$ for all ψ, and the crossover is exact.

By contrast, the *mean* does not split the distribution 50-50. For a right-skewed distribution like the Gamma:

$$P(\tilde{\varepsilon} > 1/\beta) = P\big(\mathrm{Gamma}(\psi\alpha, \beta) > \mathrm{mean}\big)$$

ranges from $e^{-1} \approx 0.368$ (exponential, $\psi\alpha = 1$) to 0.5 (Gaussian limit, $\psi\alpha \to \infty$). The CDF at the mean is *not* ψ-independent — so $D_0 = 1/\beta$ is *not* the exact crossover for step-function detection.

19. Why the mean appears in the Taylor expansion

The current result uses the cumulant expansion of $\mathbb{E}[\tilde{S}(\tilde{\varepsilon})]$ around the mean of $\tilde{\varepsilon}$:

$$\mathbb{E}[\tilde{S}(\tilde{\varepsilon})] = \tilde{S}(\mu) + \underbrace{\tilde{S}'(\mu) \cdot 0}_{\text{1st central moment}} + \frac{1}{2}\tilde{S}''(\mu)\,\sigma^2 + \frac{1}{6}\tilde{S}'''(\mu)\,\kappa_3 + \cdots$$

where $\mu = 1/\beta$, $\sigma^2 = \psi\alpha/\beta^2$, and $\kappa_3 = 2\psi\alpha/\beta^3$ is the third central moment. The mean is the natural expansion point because the first-order term vanishes, making the **second-order (variance) term the leading ψ-dependent contribution**.

Taking the ψ-derivative:

$$\frac{\partial}{\partial\psi}\,\mathbb{E}[\tilde{S}(\tilde{\varepsilon})] = \frac{\alpha}{\beta^2}\left[\frac{1}{2}\,\tilde{S}''\!\left(\frac{1}{\beta}\right) + \frac{1}{3\beta}\,\tilde{S}'''\!\left(\frac{1}{\beta}\right) + \cdots\right]$$

The leading term gives the direction rule from Section 11: $\mathrm{sign} = \mathrm{sign}(\tilde{S}''(1/\beta))$. But the **third-order correction** (weighted by the skewness $\kappa_3/\sigma^3 = 2/\sqrt{\psi\alpha}$) shifts the effective crossover.

20. The skewness correction: from mean to median

Setting the full ψ-derivative to zero (including the third-order term) gives the crossover condition:

$$\frac{1}{2}\,\tilde{S}''\!\left(\frac{1}{\beta}\right) + \frac{1}{3\beta}\,\tilde{S}'''\!\left(\frac{1}{\beta}\right) \approx 0$$

For a sigmoid-like $\tilde{S}$ transitioning at some point D_0, the second-order-only condition $\tilde{S}''(1/\beta) = 0$ gives $D_0 = 1/\beta$. Including the third-order term shifts the solution to a smaller value. For a sharp sigmoid, the corrected crossover converges to:

$$D_0^* \;\to\; \frac{2}{3\beta} \qquad \text{(the median–mode gap)}$$

recovering the exact step-function result. The gap between the two reference points is:

$$\underbrace{\frac{1}{\beta}}_{\text{mean–mode}} - \underbrace{\frac{2}{3\beta}}_{\text{median–mode}} = \frac{1}{3\beta}$$

This is precisely **one-third of the mean–mode gap**, consistent with Pearson's classical relation:

$$\text{mean} - \text{mode} \approx 3\,(\text{mean} - \text{median})$$

which gives $\text{median} - \text{mode} \approx \frac{2}{3}(\text{mean} - \text{mode})$.

21. Unified picture

> The three reference points of the Gamma profile, in mode-relative coordinates:
>
> $$\underbrace{0}_{\text{mode}} \;<\; \underbrace{\frac{2}{3\beta}}_{\substack{\text{median}\\ \text{exact crossover}}} \;<\; \underbrace{\frac{1}{\beta}}_{\substack{\text{mean}\\ \text{Jensen reference}}}$$
>
> For any detection survival function $\tilde{S}$:
>
> - **Exact crossover** (step-function $\tilde{S}$): punctuation helps detection iff $D_0 < \frac{2}{3\beta}$. This is

where $P(\tilde{\varepsilon} > D_0) = \frac{1}{2}$, independent of ψ.

- **Second-order direction rule** (smooth $\tilde{S}$): punctuation helps detection iff $\tilde{S}''\left(\frac{1}{\beta}\right) > 0$ (convex at the mean). This is a quick diagnostic that gets the *direction* right for smooth $\tilde{S}$, but slightly overestimates the "neutral zone."
- **Third-order refinement**: including the Gamma's right-skewness pulls the effective evaluation point from $\frac{1}{\beta}$ toward $\frac{2}{3\beta}$, sharpening the prediction.

Practical rule: For right-skewed infectiousness profiles (the biologically realistic case), the Jensen approximation at the mean is *conservative* — it predicts a slightly wider regime where punctuation helps detection than actually exists. The median gives the tighter, more accurate boundary. For our standard parameters ($\beta = 2$): the mean gives 0.5 days; the median gives 0.33 days; the true crossover is in between, closer to the median.

22. When does the distinction matter?

The gap between mean and median ($1/(3\beta) = T/(3\alpha)$) is small when α is large (sharp generation interval). For our parameters ($T = 5, \alpha = 10, \beta = 2$):

$$\frac{1}{3\beta} = \frac{1}{6} \text{ day} \approx 4 \text{ hours}$$

So the mean–mode diagnostic ($1/\beta = 0.5$ days) and the exact median–mode crossover ($2/(3\beta) = 0.33$ days) differ by only 4 hours. For a smooth detection survival $\tilde{S}$ that transitions over a time scale much longer than 4 hours, evaluating $\tilde{S}''$ at either point gives essentially the same answer. The distinction only matters when $\tilde{S}$ changes rapidly near the mode — i.e., when the detection process is sharp enough to be well-approximated by a step function, which is exactly when the median result becomes the right one to use.

In other words: **the two reference points are complements, not competitors.** Use the mean ($1/\beta$) for smooth $\tilde{S}$ where the curvature diagnostic is natural. Use the median ($2/(3\beta)$) for sharp $\tilde{S}$ where the step-function crossover is the right framework. They agree within $T/(3\alpha)$ days.

Step functions vs smooth $\tilde{S}$: which is the right framework?

2026-03-23

> *At the individual level, detection happens at a specific time — after which the person isolates (perfectly, for now). So the individual's "survival function" is literally a step function: $S_i(\tau) = \mathbf{1}[\tau < \tau_{\text{det},i}]$. The smooth $\tilde{S}$ is an average over the randomness in detection timing. Which framework should we use?*

23. The two representations are mathematically equivalent

For individual i with mode-anchored detection time $D_{0,i}$ (relative to their mode), the escaped transmission is:

$$J_i = R_0 \int_{\tilde{\varepsilon} < D_{0,i}} \tilde{f}_\psi(\tilde{\varepsilon})\, d\tilde{\varepsilon} = R_0\, \tilde{F}_\psi(D_{0,i})$$

The population average over the randomness in D_0 is:

$$\bar{J} = R_0\, \mathbb{E}_{D_0}\big[\tilde{F}_\psi(D_0)\big] = R_0 \int \tilde{F}_\psi(d)\, g(d)\, dd \qquad \textbf{[Step-function form]}$$

where g is the density of D_0. Alternatively, defining the smooth survival function $\tilde{S}(t) = P(D_0 > t) = 1 - G(t)$ and integrating against the infectiousness profile:

$$\bar{J} = R_0\, \mathbb{E}_{\tilde{\varepsilon}}\big[\tilde{S}(\tilde{\varepsilon})\big] = R_0 \int \tilde{S}(t)\, \tilde{f}_\psi(t)\, dt \qquad \textbf{[Smooth-}\tilde{S}\textbf{ form]}$$

These are **identical by Fubini's theorem**:

$$\mathbb{E}_{\tilde{\varepsilon}}[\tilde{S}(\tilde{\varepsilon})] = \mathbb{E}_{\tilde{\varepsilon}}\big[P(D_0 > \tilde{\varepsilon})\big] = \mathbb{E}_{\tilde{\varepsilon}}\big[\mathbb{E}_{D_0}[\mathbf{1}_{D_0 > \tilde{\varepsilon}}]\big] = \mathbb{E}_{D_0}\big[\mathbb{E}_{\tilde{\varepsilon}}[\mathbf{1}_{\tilde{\varepsilon} < D_0}]\big] = \mathbb{E}_{D_0}\big[\tilde{F}_\psi(D_0)\big]$$

So the two representations always give the same numerical answer. The choice between them is

about **which source of randomness to average over first**.

24. Why the step-function framework is more fundamental here

Although mathematically equivalent, the step-function / D_0 representation has three advantages for understanding the ψ-effect:

The escaped transmission depends on the interaction of two random variables:

- $\tilde{\varepsilon} \sim \tilde{f}_\psi$: **when** an individual transmits (relative to their mode). This is the biological source of randomness, controlled by ψ.
- $D_0 \sim g$: **when** an individual is detected (relative to their mode). This is the detection source of randomness, independent of ψ.

The step-function framework keeps these separate: $\tilde{F}_\psi(D_0)$ is the biological quantity, and averaging over g adds the detection noise. The smooth-$\tilde{S}$ framework pre-averages over D_0, collapsing the detection randomness into the shape of $\tilde{S}$.

In the step-function framework, for each realisation of D_0, there is a clean crossover at $D_0^* = \text{median} - \text{mode} \approx 2/(3\beta)$: for $D_0 < D_0^*$, spikes escape more; for $D_0 > D_0^*$, smooth escapes more. The population-level ψ-effect is simply a matter of **how much of the D_0 distribution falls on each side**:

$$\underbrace{\mathbb{E}_{D_0}[\tilde{F}_\psi(D_0)]}_{\text{escaped fraction}} \quad \text{favours spikes} \iff \text{most of } g(D_0) \text{ is to the left of } D_0^*$$

This is immediate and visual. The smooth-$\tilde{S}$ framework gives the same answer via the curvature diagnostic $\tilde{S}''(1/\beta)$, but that's a less transparent encoding of the same fact.

The earlier analysis (Sections on "Random detection times") showed that noise in D_0 shifts the effective crossover *rightward*:

$$D_0^{*,\text{eff}} \approx \frac{2}{3\beta} + c\,\sigma_{D_0}^2$$

This is a statement about the distribution of D_0, not about the shape of $\tilde{S}$. It says: even if the average detection time is at the nominal crossover, randomness in detection timing tips the balance toward spikes (because spikes benefit asymmetrically from lucky early detections). In the smooth-$\tilde{S}$ framework, this result is buried in the curvature of $\tilde{S}$ and harder to interpret.

25. When is the smooth-$\tilde{S}$ framework useful?

The smooth-$\tilde{S}$ representation is still valuable in specific situations:

- **Parametric convenience:** If the detection process gives $\tilde{S}$ a nice closed form (exponential, Weibull, etc.), the curvature diagnostic $\tilde{S}''(1/\beta)$ is a quick one-line computation.
- **Complex detection processes:** When D_0 doesn't have a simple distribution (e.g., it depends on multiple competing processes — symptom onset, scheduled tests, contact tracing — and isolation occurs at the first trigger), it may be easier to characterise the resultant $\tilde{S}$ directly than to derive the distribution of D_0.
- **Imperfect isolation:** If isolation is not all-or-nothing (e.g., transmission is reduced but not eliminated), the individual response is no longer a step function. Then $\tilde{S}$ becomes the natural framework, encoding both the timing and the degree of isolation.

26. Synthesis: the D_0 framework as the "master" approach

> **For mode-anchored detection with perfect isolation:**
>
> The step-function / D_0 framework is the more fundamental representation. The escaped transmission is:
>
> $$R_{\text{eff}} = R_0\,\mathbb{E}_{D_0}\big[\tilde{F}_\psi(D_0)\big]$$

> The ψ-effect is determined by where the distribution of D_0 sits relative to the crossover $D_0^* \approx 2/(3\beta)$:
>
> - $\mathbb{E}[D_0] \lesssim 2/(3\beta) + c\,\sigma_{D_0}^2$: spikes escape more (punctuation helps detection)
> - $\mathbb{E}[D_0] \gtrsim 2/(3\beta) + c\,\sigma_{D_0}^2$: smooth escapes more (punctuation hurts detection)
>
> The smooth-$\tilde{S}$ representation ($\tilde{S}(t) = P(D_0 > t)$) gives the same answer via $\tilde{S}''(1/\beta)$, and is useful as a computational shortcut or when detection is complex. But the conceptual content lives in the D_0 framework: **the median–mode gap is the crossover, and detection noise favours spikes.**
>
> The smooth-$\tilde{S}$ framework becomes the natural one when we relax perfect isolation — then the individual response is no longer a step function and $\tilde{S}$ genuinely describes a graded process.